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# A Unified Convergence Theory for Non-Stationary Reinforcement Learning

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Anonymous Author(s)

Affiliation

Address

email

## Abstract

Non-stationary reinforcement learning lacks a unified convergence theory: variation-budget regret bounds, sliding-window methods, tempo-and-forgetting analyses, two-term decompositions, and causal-RL frameworks each address fragments without composing. We present a unified theory through the composition of four structural elements: (i) a two-gap diagnostic separating goal-feasibility from policy-quality; (ii) a point-mass reverse-KL/TV identity under deterministic optimum,  $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}}$ , an *exact* algebraic identity that lies strictly below the Pinsker and Bretagnolle–Huber envelopes at this deterministic corner (collapsing the BH inequality’s envelope to a single value rather than introducing a new bounding technique); (iii) a multi-factor strategic tempo  $\mathcal{T}_\Sigma = \min_{(i,j)} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$  with a structural forgetting prerequisite  $\mathcal{T}_\Sigma > \rho_\Sigma / R_\Sigma$  as a survival inequality; (iv) closed-loop interventional access making regret bounds learnable from on-policy data via the Pearl causal hierarchy. Composing these gives a cumulative dynamic regret theorem with rate  $O(V_{\max} \sqrt{(B_T + 1)T})$  where  $B_T$  counts optimum-change events. We further establish a structural failure class: every gain-decay update (effective per-element gain  $\alpha_{ij}^{(t)} \rightarrow 0$ , including count-accumulating Bayesian schemes and Robbins–Monro gradient methods) eventually violates the forgetting prerequisite for any positive disturbance rate, while finite-gain mechanisms admit bidirectional thresholds (constant-step stochastic approximation, sliding windows, bounded memory, block restart). The point-mass identity is coordinate-optimal among bounds depending only on total variation, chain-rule compositional to multi-step trajectory regret as behavior-cloning loss against the optimal trajectory, and load-bearing for the cumulative theorem. The deterministic-optimum scope extends perturbatively to  $\epsilon$ -stochastic and softmax-regularized policies with  $O(\epsilon \log(1/\epsilon))$  and  $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$  corrections respectively (exponentially small with polynomial prefactor). Theory-only; Lemma 5.2 makes the ProST reduction rigorous at the sector-parameter level, while their published experiments motivate the strategic-tempo component.

## 1 Introduction

Non-stationary reinforcement learning has matured along four largely separate research tracks. Each has produced rigorous results; none has been composed with the others into a single convergence theory.

The four tracks are visible in the recent literature as parallel lineages. **Variation-budget dynamic regret under drift**<sup>[1–4]</sup> gives sublinear dynamic regret under bounded total variation in reward and transition dynamics. **Two-term regret decompositions**<sup>[5–7]</sup> split the dynamic regret along an *exploration-vs-adaptation* axis, isolating the cost of confidence-set construction from the cost of

tracking a moving optimum. **Tempo and forgetting analyses**<sup>[8–12]</sup> make policy-update timing or evidence-discount rate an explicit convergence variable. **Causal and interventional access**<sup>[13–18]</sup> uses causal-graph structure to sharpen sample complexity, but operates in stationary settings only.

The information-theoretic regret literature<sup>[19–24]</sup> traverses these tracks at a different layer, using entropy, mutual information, information ratio, Pinsker, or Hellinger to bound regret. Notably absent from that literature, as we document below (Appendix F), is an *exact* reverse-KL / total-variation identity at the deterministic-optimum corner: a sharp action-distribution regret coordinate that strictly improves both Pinsker and the Bretagnolle–Huber inequality<sup>[25]</sup> at this point. The Bretagnolle–Huber inequality is itself absent from the retrieved RL/non-stationary corpus *as an upper-bound regret coordinate* (its use in lower-bound proofs via Le Cam’s method<sup>[26,27]</sup> is well established and not contested here); the *exact identity at its point-mass corner* — which we establish — is *a fortiori* absent.

These four tracks share an evident common ancestor — the dynamic-regret analysis of online MDPs — but no published framework composes them. In particular, no framework combines (a) a regret decomposition along the *goal-feasibility-vs-policy-quality* axis (distinct from the exploration-vs-adaptation decompositions); (b) an exact point-mass reverse-KL/TV regret identity under deterministic optimum, strictly tighter than the Bretagnolle–Huber inequality at this corner; (c) a structural survival inequality threading rate-of-policy-revision against environment-side disturbance; and (d) a closed-loop causal-access argument that makes the regret bound *learnable* on-policy rather than merely provable analytically. The fundamental question remains open:

*Can we design a unified convergence theory for non-stationary reinforcement learning that handles non-stationarity, has explicit metric structure on policy space, and is learnable from on-policy data?*

In this paper we present the first composition that delivers all three properties jointly. Composing four structural elements — a two-gap diagnostic, a point-mass reverse-KL/TV identity, a multi-factor strategic tempo with forgetting prerequisite, and closed-loop interventional access — yields a cumulative dynamic regret theorem with rate  $\text{DynReg}(T) \leq \tilde{O}(N_h \sqrt{(B_T + 1)T})$  in the piecewise-stationary regime, where  $B_T$  counts piecewise-stationary segment boundaries (kernel-and-reward changes) along the realized trajectory and  $N_h$  is the planning horizon. The per-round coordinate of this rate is sharper than both Pinsker and Bretagnolle–Huber at the deterministic- $\pi^*$  corner. The technical anchor is an exact algebraic identity:  $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}$ , which holds for any policy  $Q$  with  $Q(a^*) > 0$  and lies *strictly below* both the Pinsker envelope  $\sqrt{D_{\text{KL}}/2}$  and the Bretagnolle–Huber envelope  $\sqrt{1 - e^{-D_{\text{KL}}}}$  at this corner.

**Roadmap** Section 2 situates the four-track landscape. Section 3 sets minimal preliminaries. Section 4 presents the four-component composition theorem with narrative unpacking and a defense of why each component is necessary. Section 5 sketches the proof via four key lemmas chained through a block decomposition. Section 6 gives concluding observations and practitioner takeaways. Full proofs and auxiliary mathematical material are in the appendices.

## 1.1 Contribution

The contribution has the shape — recognized by the NeurIPS Theory Track guidelines as a valid form of originality — of “*a novel combination of existing techniques [where] the reasoning behind this combination is well-articulated.*” Three specific moves carry the paper.

**(i) Point-mass reverse-KL/TV regret identity (Section 4, full development Section 5.1).** Under deterministic optimum policy  $\pi^* = \delta_{a^*}$ , an elementary calculation gives the *exact identity*  $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q))$ , equivalently  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$ . Substituting this into the classical Bretagnolle–Huber inequality  $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$  yields  $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ , which holds *strictly* on  $D_{\text{KL}} > 0$  since  $x < \sqrt{x}$  on  $(0, 1)$ . At the deterministic- $\pi^*$  corner the identity supplies the *exact* TV value strictly below the BH envelope, positioning BH as the lineage the identity sits within rather than as the source it instantiates. Composed with the textbook total-variation regret bound (full development Section 5.1), the identity yields a tight two-sided characterization  $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)}) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)})$ , *strictly below* the Pinsker form  $V_{\max} \sqrt{D_{\text{KL}}/2}$  — at  $D_{\text{KL}} = 0.01$  the identity bound is  $7\times$  tighter; at

88  $D_{\text{KL}} > 2$  Pinsker becomes vacuous against the trivial  $V_{\text{max}}$  envelope while the identity continues to  
 89 saturate at  $V_{\text{max}}$ . The deterministic- $\pi^*$  scope is not a hard wall: the identity extends *perturbatively* to  
 90  $\epsilon$ -greedy and softmax-regularized optima with quantified  $O(\epsilon \log(1/\epsilon))$  and  $O(\tau^{-1} \exp(-\Delta_{\text{min}}/\tau))$   
 91 corrections respectively (full development in Section 5.1 and Appendix B.2), recovering exact equality  
 92 at  $\epsilon = 0 / \tau \rightarrow 0$ .

93 **(ii) Strategic tempo with forgetting prerequisite (Section 4, full development Section 5.2).** A  
 94 multi-factor strategic tempo  $\mathcal{T}_{\Sigma}$  — composed per revisable policy element of an observation rate  
 95  $\nu$ , an identifiability coefficient  $\iota$ , and an update gain  $\eta$  — yields a structural survival inequality  
 96  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_{\Sigma}/R_{\Sigma}$  in which all three factors are multiplicatively  
 97 load-bearing per element, with the *bottleneck* (per-element minimum) as aggregator. The familiar  
 98 single-factor form  $(1 - \lambda) > \rho_{\Sigma}/R_{\Sigma}$  is recovered as a corollary under uniform-Regime-A Beta-  
 99 Bernoulli normalization. A structural-class theorem identifies the *gain-decay* class  $\mathcal{A}_{\text{decay}}$  (count-  
 100 accumulating Bayesian updates, observation-aggregating schemes without restart, gradient-based  
 101 Robbins–Monro decaying-step methods) as one in which every agent eventually violates the threshold  
 102 for any positive disturbance rate, while non-accumulating mechanisms (constant-step SA, sliding-  
 103 window, bounded-memory, block-restart) face *bidirectional* ceilings. This *lifts* the tempo result of<sup>[8]</sup>  
 104 from a single-factor hyperparameter-optimization claim to a multi-factor structural-threshold claim.  
 105 The forgetting prerequisite is the strategic analog of the persistence condition in adaptive control<sup>[28]</sup>,  
 106 with  $(1 - \lambda)$  playing the role of adaptive tempo per element.

107 **(iii) Closed-loop interventional access (Section 4, full development Section 5.3).** With explicit  
 108 grounding in Bareinboim’s causal-hierarchy theorem<sup>[29]</sup>, the agent in the feedback loop *generates*  
 109 Pearl Level-2 (interventional) data by construction: the action  $a_t$  causally precedes  $o_{t+1}$ , so under  
 110 positivity, sequential ignorability, and a known action mechanism, the loop’s data-generating process  
 111 is interventional rather than associational. This makes the regret bound of (i) *learnable* from on-policy  
 112 interaction in regimes with sufficient identifiability, not merely provable in principle. The argument is  
 113 implicit in the action-perception-loop framing of active inference<sup>[30,31]</sup> and the causal-RL line<sup>[13–18]</sup>;  
 114 we make the Bareinboim-hierarchy connection explicit, partition usable strength of identification  
 115 into three regimes (A: full; B: partial; C: observation-only), and flag honestly where the loop yields  
 116 interventional *data* without yielding identified *do*-estimates.

117 **Connective tissue: the two-gap diagnostic (Section 4).** A separate decomposition of the regret  
 118 signal — the *satisfaction gap*  $\delta_{\text{sat}}$  (the goal is unmet under the best available one-step improvement)  
 119 versus the *control regret*  $\delta_{\text{regret}}$  (the gap between the best available improvement and the agent’s current  
 120 policy) — provides the corrective-action routing that ties (i)–(iii) together. The  $2 \times 2$  disambiguation  
 121 routes four regimes (success, strategy problem, capability limit, both) to four distinct corrective  
 122 actions, along an axis orthogonal to the exploration-vs-adaptation decompositions of<sup>[5–7]</sup>.

123 **Composition** The four together yield (Section 4) a non-stationarity-aware convergence theory with  
 124 three properties no existing strand exhibits jointly: the bound *handles non-stationarity* (via tempo and  
 125 forgetting); has *explicit metric structure on policy space* (via the point-mass reverse-KL identity); is  
 126 *learnable from on-policy data* (via closed-loop interventional access). Composition with a variation-  
 127 budget block argument under a restart-on-change base learner yields the cumulative dynamic regret  
 128 bound stated above — sharper per-round coordinate than Pinsker/BH, with the cumulative rate sitting  
 129 in the piecewise-stationary  $B_T$  family of<sup>[1]</sup> (*not* directly comparable to the continuous-variation  
 130 rate  $\tilde{O}(SA V_T^{1/3} H T^{2/3})$  of<sup>[3]</sup>; the two regimes coincide up to constants under abrupt-magnitude  
 131 piecewise-stationarity). Each of the four components is itself a theorem (cited or proved); the paper’s  
 132 contribution is the recognition that the four together close a story none of the strands closes individually,  
 133 plus the cumulative-regret rate that follows from composing the per-round identity with a base-learner  
 134 stationary rate.

## 135 1.2 Scope and limitations

136 **What “convergence” means here** We do not claim pointwise policy convergence — under genuine  
 137 non-stationarity the optimum is itself moving. By *convergence* we mean: per-round regret-coordinate  
 138 convergence within stationary blocks; cumulative dynamic regret with sublinear / vanishing-average  
 139 rate when  $B_T = o(T)$  (Cesàro tracking  $\frac{1}{T} \sum_t (V_t^* - V(\pi_t)) \rightarrow 0$ ); and ultimate boundedness of  
 140 the strategic mismatch  $\|\delta_{\Sigma}\|$  under the tempo threshold. Pointwise  $V(\pi_t) \rightarrow V^*$  is structurally  
 141 unavailable under genuine non-stationarity; the Cesàro tracking statement is the right replacement.

**Theory only** No experiments. The point-mass identity is mathematically airtight (a two-line direct calculation that specializes the BH family at its deterministic- $\pi^*$  corner). The empirical grounding for the strategic-tempo claim is provided indirectly through reduction to<sup>[8]</sup> ProST as a special case (Section 5); at the steady-state sector-parameter level the reduction is rigorous.

**Canonical scope** The composition theorem holds under deterministic optimum policy, bounded value range, isolated optimum (so  $\Delta_{\min} > 0$ ), and a singular causal trajectory in the sense made precise in Section 3. The deterministic- $\pi^*$  scope extends *perturbatively* to  $\epsilon$ -stochastic and softmax-regularized optima (Section 5.1); tied-optimum is a degraded one-sided form (Appendix C.5). Each axis of the canonical scope degrades cleanly outside it (we discuss the failure modes in Section 6).

**Comparator regime** The cumulative rate is in the piecewise-stationary  $B_T$  family, not the continuous-variation  $V_T$  family of<sup>[3]</sup>. Generalizing the per-round-identity route to continuous variation is open (Section 6).

## 2 Related Work

The four tracks named in Section 1 organize the literature.

**Variation-budget dynamic regret** <sup>[1–4,32]</sup> establish dynamic regret bounds under a *variation budget*  $V_T$  on cumulative reward and transition variation.<sup>[3]</sup> achieves the near-optimal continuous-variation rate  $\tilde{O}(SA V_T^{1/3} H T^{2/3})$  via RestartQ-UCB;<sup>[1]</sup> treats the piecewise-stationary case with restart-on-change. Our cumulative rate  $\tilde{O}(N_h \sqrt{(B_T + 1)T})$  sits in the piecewise-stationary family — same block-shape as Cheung-style restart-on-change — recovered as a corollary of composition with the per-round identity. The novelty is the *threshold form* of the strategic-tempo inequality (environment regimes where no  $\lambda \in (0, 1)$  stabilizes the sector-Lyapunov mismatch), which dynamic-regret-optimization analyses do not surface.

**Two-term regret decompositions along the exploration-vs-adaptation axis** <sup>[5–7]</sup> decompose the dynamic regret into a *confidence-set construction* term and an *adaptation-under-non-stationarity* term.<sup>[33]</sup> treats a structurally adjacent question — constraint-region feasibility — using a two-term decomposition along that axis. Same shape, different axis: ours is goal-feasibility-vs-policy-quality (the satisfaction gap distinguishes “the goal is unattainable from  $M_t$ ” from “the current policy is suboptimal”); theirs is uncertainty-source. Neither subsumes the other: a Long-Fei Li-style exploration term can be present in any of our  $2 \times 2$  cells, and an adaptation term likewise. The two axes are orthogonal.

**Tempo and forgetting** <sup>[8]</sup> (ProST) defines an agent tempo as the schedule of policy update times  $\{t_1, \dots, t_K\}$  and computes the schedule that minimizes a dynamic regret upper bound.<sup>[9]</sup> (Pausing Policy Learning) refines this with non-zero hold durations.<sup>[10–12]</sup> establish forgetting / sliding-window analyses for non-stationary linear MDPs and bandits. We *lift* the ProST move along two axes simultaneously: single-factor  $\rightarrow$  multi-factor (per-element  $\nu \cdot \iota \cdot (1 - \lambda)$  rather than scalar update frequency); and hyperparameter-optimization  $\rightarrow$  structural-survival inequality (the threshold below which no schedule persists, irrespective of which schedule is chosen). ProST recovers as the Beta-Bernoulli special case  $|E| = 1, \nu = \iota = 1$ .

**Causal and interventional access** <sup>[13–18,34]</sup> use causal-graph structure to sharpen sample complexity in RL. Their setting is *stationary*. We compose the closed-loop interventional-access argument with non-stationarity. The connection between Pearl’s causal hierarchy<sup>[35]</sup> and the data-generating process of an agent in a feedback loop<sup>[29]</sup> is the technical bridge.

**Information-theoretic regret (cross-cutting)** <sup>[19–24,36]</sup> traverse the above tracks at a different layer, using entropy / mutual information / information ratio / Pinsker / Hellinger as the divergence backbone. The Russo–Van Roy information ratio uses Pinsker as its Cauchy–Schwarz follow-up; the IDS line and its descendants use mutual-information bounds. The Bretagnolle–Huber inequality<sup>[25]</sup> — well-known in lower-bound proofs via Le Cam’s method<sup>[26,27]</sup> — does not appear *as an upper-bound regret coordinate* in this corpus, and its exact point-mass corner (our Lemma 5.1) is *a fortiori* absent. We document the retrieval scope and methodology in Appendix F.

191 **Adjacent (satisficing / feasibility)** <sup>[37,38]</sup> use *satisficing* to mean “any policy above acceptance  
 192 level  $\beta$  is acceptable.” Our  $\delta_{\text{sat}}$  is structurally distinct:  $V_O^{\min}$  is a property of the objective (set by  
 193 the application domain), and  $\delta_{\text{sat}} > 0$  signals goal-relative *infeasibility* (no policy in  $\Pi$  meets the  
 194 threshold), not policy-relative satisficing.

195 **Contemporaneous (post-March 2026)** <sup>[38,39]</sup>; cited and distinguished without empirical-  
 196 comparison demand per the contemporaneous-work cutoff convention.

### 197 3 Preliminaries

198 We adopt strict minimalism: only the notation and assumptions used directly in the main result and  
 199 the mechanism statements appear here. Convention-hierarchy details (one-step vs. receding-horizon  
 200 vs. Bellman continuation), the action-gap matching argument, tied-optimum extensions, and the  
 201 deterministic-trajectory deployment modes are deferred to Appendix C.

202 **Episodic non-stationary MDP** A finite-horizon non-stationary MDP  $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$  with state  
 203 space  $\mathcal{S}$ , finite action space  $\mathcal{A}$ , time-indexed transition kernels  $P_t(\cdot \mid s, a)$  and reward functions  
 204  $r_t(s, a)$  allowed to vary across rounds  $t$ , and finite planning horizon  $N_h$ . Standard non-stationary-RL  
 205 boundedness assumptions<sup>[1]</sup>: per-step reward in  $[0, 1]$ .

206 **Variation regimes** A variation budget  $V_T$  measures continuous variation in transitions and rewards.  
 207 We work in the *piecewise-stationary* specialization: time partitions into  $B_T + 1$  maximal stationary  
 208 intervals on which  $(P_t, r_t)$  is constant, with  $B_T := |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$  counting *kernel-*  
 209 *stationary-segment boundaries*. Optimum-changes  $a_t^* \neq a_{t-1}^*$  can only occur at segment boundaries  
 210 (since  $a^*$  is determined by  $(P_t, r_t)$ ); the converse is not generally true (a segment boundary need not  
 211 change  $a^*$  — e.g., uniform reward shifts that lift all actions, or kernel shifts that preserve the argmax).  
 212 Counting kernel-segment boundaries rather than optimum-change events is the right measure for the  
 213 base learner’s per-block stationary-regret guarantee in (A5): UCRL2 / UCBVI / Thompson sampling  
 214 build empirical confidence sets on  $(P, r)$ , and within-block kernel drift breaks the per-block  $\sqrt{L}$  rate  
 215 even when  $a^*$  is unchanged. In the bandit case with isolated optimum and reward shifts that always  
 216 cross argmax,  $B_T$  reduces to the optimum-change count  $|\{t : a_t^* \neq a_{t-1}^*\}|$  — a useful corollary in  
 217 regime ARG (Section 4 *Comparator regime*) but not the headline framing. The two variation regimes  
 218 ( $B_T$  vs. continuous  $V_T$ ) are distinct in general; under abrupt-magnitude- $\delta$  piecewise-stationarity they  
 219 coincide up to constants ( $V_T \asymp \delta B_T$ ). Our results are stated for  $B_T$ ; the continuous- $V_T$  extension is  
 220 open (Section 6).

221 **Policy distribution** The agent’s policy at round  $t$  is a distribution  $Q_t(\cdot \mid s)$  over actions; we write  $Q$   
 222 when state and round are understood. We adopt the canonical scope: the optimum policy at  $(M_t, s)$   
 223 is *deterministic*,  $\pi^*(\cdot \mid M_t, s) = \delta_{a^*(s)}$  where  $a^*(s) := \arg \max_a Q^\pi(s, a)$  is the optimal action  
 224 under the agent’s current model  $M_t$ . Deterministic  $\pi^*$  is canonical for finite-MDP RL with isolated  
 225 optima<sup>[27]</sup>; the perturbative extension to  $\epsilon$ -stochastic and softmax-regularized optima (Section 5.1,  
 226 full derivation Appendix B.2) covers smooth deviations.

227 **Value object** Given an objective  $O$ , model  $M_t$ , policy  $\pi$ , and horizon  $N_h$ :  $Q_O(M_t, a; \pi_{\text{cont}}, N_h) =$   
 228  $\mathbb{E}[V_O(\tau) \mid M_t, \text{do}(a_t = a), a_{t+1} \sim \pi_{\text{cont}}]$ . The  $\text{do}(\cdot)$ <sup>[35]</sup> flags intervention on  $a_t$ , not conditioning.  
 229 We adopt the **one-step improvement** convention  $\pi_{\text{cont}} = \pi_{\text{current}}$  as default: most conservative, compa-  
 230 rable across rounds, fixed-point-free. Receding-horizon and Bellman alternatives are in Appendix C.1.

231 **Bounded value range**  $V_{\max}(M_t) := \max_a Q_O(M_t, a) - \min_a Q_O(M_t, a)$ , finite under bounded  
 232 reward and finite horizon.

233 **Action gap (isolated optimum)**  $\Delta(a) := Q_O(a^*) - Q_O(a) \in [0, V_{\max}]$ ,  $\Delta_{\min} := \min_{a \neq a^*} \Delta(a) >$   
 234  $0$  whenever the optimum is isolated.

235 **Strategic regret** For policy distribution  $Q$ :  $R(Q) := Q_O(a^*) - \mathbb{E}_{a \sim Q}[Q_O(a)] = \sum_{a \neq a^*} Q(a) \cdot$   
 236  $\Delta(a)$ . Three classical forms —  $\mathbb{E}_{\pi^*}[V] - \mathbb{E}_Q[V]$ ,  $V(a^*) - \mathbb{E}_Q[V]$ ,  $\mathbb{E}_{\pi^*}[V - V_Q]$  — coincide under  
 237 deterministic  $\pi^*$ .

238 **Total variation and reverse-KL** For finite action measures  $P, Q$  on  $\mathcal{A}$ ,  $\text{TV}(P, Q) :=$   
239  $\frac{1}{2} \sum_a |P(a) - Q(a)|$ ,  $D_{\text{KL}}(P \| Q) := \sum_a P(a) \log \frac{P(a)}{Q(a)}$  with the convention  $0 \log 0 = 0$ .  
240 Our bounds use the *reverse* KL direction (with  $P = \pi^*$ ); the forward direction  $D_{\text{KL}}(Q \| \pi^*)$  is  $+\infty$   
241 whenever  $Q$  has off-optimum mass and is therefore vacuous as a regret coordinate (full direction-  
242 forcing argument in Appendix C.3).

243 **Singular trajectory** The agent operates on a single, non-forkable causal trajectory:  $a_t$  causally  
244 precedes  $o_{t+1}$ , and the model state  $M_t$ ’s sufficiency is defined relative to *this* trajectory rather than to  
245 a model-state equivalence class. This is the operational ground for the do-operator above and for the  
246 closed-loop interventional-access argument (Section 5.3).

## 247 4 Main Result

248 We present the four components first as motivation, then state the formal composition theorem, then  
249 unpack what its conclusions mean.

### 250 4.1 The four components

251 Each component supplies a *distinct epistemic property* that the others cannot deliver. The composition  
252 is necessary for the joint properties (rate + persistence + learnability + corrective-action routing), not  
253 for the rate alone. The component-by-component failure-mode analysis is in Section 4.4 below.

254 **Component 1: Two-gap diagnostic — corrective-action routing** Let  $\Pi$  denote the policy class  
255 available to the agent and let  $V_O^{\min}$  denote the threshold trajectory value above which the objective is  
256 “met.” Define *objective attainability*  $A_O(M_t; \Pi, N_h) := \sup_{\pi \in \Pi} V_O(M_t, \pi; N_h)$ , the *satisfaction*  
257 *gap*  $\delta_{\text{sat}} := V_O^{\min} - A_O$  (goal-feasibility diagnostic; positive means *unmet* under the best available  
258 policy), and the *control regret*  $\delta_{\text{regret}} := A_O - V_O(M_t, \pi_{\text{current}}; N_h) \geq 0$  (policy-quality diagnostic).  
259 The  $2 \times 2$  cross of these two gaps routes corrective action: *Success* (no action), *Strategy problem*  
260 (revise the policy), *Capability limit* (revise the model, policy class, or horizon — in that order), *Both*  
261 (revise the policy first, then re-evaluate). Without this routing, an agent receiving a regret signal  
262 cannot distinguish “the goal is unattainable from  $M_t$ ” from “the current policy is suboptimal.” A  
263 single  $\delta_{\text{objective}} = V_O^{\min} - V_O(M_t, \pi_{\text{current}}; N_h)$  collapses both regimes into a single number and loses  
264 the diagnostic. The two-gap split runs along an axis *orthogonal* to the exploration-vs-adaptation  
265 decompositions of<sup>[5–7]</sup>: theirs isolates uncertainty *sources*; ours isolates *what is wrong*.

266 **Component 2: Point-mass reverse-KL/TV identity — metric layer on policy space** Under  
267 deterministic  $\pi^* = \delta_{a^*}$ , the reverse-KL and total variation between  $\pi^*$  and any policy  $Q$  are related by  
268 the *exact identity*  $D_{\text{KL}}(\pi^* \| Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q)) \iff \text{TV}(\pi^*, Q) =$   
269  $1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$ . This is a two-line direct calculation under the point-mass assumption (full statement  
270 and proof in Section 5.1). It is *not* the Bretagnolle–Huber inequality at equality: substituting it into  
271 the BH form  $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$  gives  $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ , *strictly* on  $D_{\text{KL}} > 0$ . The  
272 identity sits *strictly below* the BH envelope at this corner. Composed with the textbook TV-regret  
273 bound  $R(Q) \leq V_{\max} \text{TV}(\pi^*, Q)$  and the matching action-gap lower bound, it yields the two-sided  
274 characterization  $\Delta_{\min}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)}) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \| Q)})$ , *Lipschitz-*  
275 *equivalent* with constants  $\Delta_{\min}/V_{\max}$  and 1. The identity coordinate is *coordinate-optimal among*  
276 *bounds depending only on TV*: any tighter bound on  $R(Q)$  requires information beyond TV (the  
277 value-landscape spread). Strict improvement over Pinsker is uniform: at  $D_{\text{KL}} = 0.01$  the identity  
278 bound is  $7 \times$  tighter; for  $D_{\text{KL}} > 2$  Pinsker is vacuous against the trivial  $V_{\max}$  envelope while the identity  
279 saturates at  $V_{\max}$ . The deterministic- $\pi^*$  scope is not a hard wall: the identity extends *perturbatively* to  
280  $\epsilon$ -greedy and softmax-regularized optima with corrections  $O(\epsilon \log(1/\epsilon))$  and  $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$   
281 respectively — exponentially small with polynomial prefactor in the softmax case (Section 5.1).

282 **Component 3: Multi-factor strategic tempo with forgetting prerequisite — non-stationarity**  
283 **persistence** Consider a structured policy with  $|E|$  revisable elements (in our analysis, edges of  
284 an internal causal model the agent maintains over policy components; in Lee et al. 2023<sup>[8]</sup>’s ProST,  
285 sub-policies indexed by a tempo schedule). For each element  $(i, j)$ , let  $\nu_{ij}$  denote the per-element  
286 observation rate,  $\iota_{ij} \in [0, 1]$  the identifiability coefficient (fraction of evidence that genuinely identifies

the causal effect rather than reflecting confounded association), and  $\eta_{\text{edge},ij}$  the per-element update gain. Within a sector-Lyapunov reduction with diagonal correction (full Model ( $\Sigma$ ) in Section 5.2), per-element exponential forgetting at rate  $\lambda_{ij}$  gives steady-state gain  $\eta_{\text{edge},ij}^{\text{ss}} \approx 1 - \lambda_{ij}$ , and ultimate boundedness of the strategic mismatch  $\|\delta_\Sigma\|$  within the modeled dynamics is *sufficient* under the bottleneck threshold  $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$ . All three factors are multiplicatively load-bearing per element (zeroing any collapses the bottleneck; low identifiability cannot be compensated by faster forgetting), and the *bottleneck* (per-element minimum, not aggregate sum) is the right aggregator: a Lyapunov derivation under adversarial disturbance saturates at min, not  $\sum$ , when the disturbance concentrates on the weakest element. The structural-class theorem identifies the *gain-decay* class  $\mathcal{A}_{\text{decay}}$  — count-accumulating Bayesian updates without forgetting (e.g., Beta-Bernoulli with  $\eta = 1/(n+1)$ ), bounded-memory schemes with growing memory, observation-aggregating schemes without restart, *and* gradient-based methods with vanishing step sizes (the Robbins–Monro regime,  $\eta_t = c/t^\beta$  for  $\beta \in (0, 1]$ ) — as one in which *every* agent eventually violates the threshold for any positive disturbance rate (formal statement in Appendix C.6). Non-accumulating mechanisms (constant-step SA, sliding-window, bounded-memory, block-restart) face *bidirectional* ceilings: the threshold is tight in both directions and the schedule has to be in the right regime, not arbitrarily fast or slow. The familiar single-factor form  $(1 - \lambda) > \rho_\Sigma / R_\Sigma$  is recovered as a corollary under uniform-Regime-A Beta-Bernoulli normalization.

**Component 4: Closed-loop interventional access — on-policy learnability** Pearl’s causal hierarchy<sup>[35]</sup> and the causal-hierarchy theorem of<sup>[29]</sup> establish that Level-2 (interventional) queries  $P(Y \mid \text{do}(X))$  are not in general identifiable from Level-1 (associational) data  $P(Y \mid X)$ . By temporal ordering and the singular-trajectory commitment of Section 3,  $a_t$  causally precedes  $o_{t+1}$ . Under three conditions on the agent’s action mechanism — (C1) *positivity* ( $\pi_t(a \mid H_t) \geq p_{\min} > 0$  on the support of the information state  $H_t$ ), (C2) *sequential ignorability* (the information state blocks unobserved confounding of action selection), and (C3) *known action mechanism* (the behavior policy is known) — conditioning on  $H_t$  blocks the action-selection confounding pathway, so the conditional distribution  $P(o_{t+1} \mid a_t, H_t)$  coincides with the interventional distribution  $P(o_{t+1} \mid \text{do}(a_t), H_t)$ . The loop’s data-generating process is *interventional* by construction (formal statement in Section 5.3). This makes the regret bound of Component 2 *learnable on-policy* in regimes with sufficient identifiability: the agent’s identified optimum  $a_{\text{ag},t}^*$  matches the true optimum  $\bar{a}_t^*$  with probability  $p_{\text{id}}$ , and the per-state, per-step bias from misidentification is bounded by  $(1 - p_{\text{id}}(s)) \log(1/q_0)$  under two-point support  $Q_t \geq q_0$  (Section 5.4). Three regimes<sup>[29]</sup> partition usable strength: A (intervention-rich,  $\iota \approx 1$ ), B (partial,  $\iota \in (0, 1)$ ), C (observation-only,  $\iota \approx 0$ ).

## 4.2 The composition theorem

**Theorem 4.1** (Composition: joint guarantees with conditional cumulative reduction). *Let  $(\mathcal{S}, \mathcal{A}, P_t, r_t, N_h)$  be a non-stationary MDP with bounded reward, finite action space, and for each round  $t$  a deterministic optimum policy  $\pi_t^* = \delta_{a_t^*}$  with action gap  $\Delta_{\min} > 0$ . Suppose the agent operates on a singular causal trajectory in the sense of Section 3 and updates its policy with per-element exponential forgetting at rates  $\{\lambda_{ij}\}$ . Let  $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$  be the per-state reverse-KL coordinate (in the bandit case  $N_h = 1$  the state argument is suppressed). If*

*(A1) Metric extension. The identity  $\text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$  of Lemma 5.1 is read in the extended real sense: when  $Q_t(a_t^*(s) \mid s) = 0$ ,  $K_t(s) = +\infty$  and  $1 - e^{-K_t(s)} = 1$  recovers the trivial bound  $R(Q_t \mid s) \leq V_{\max}$ . No pointwise positivity is required; vanilla deterministic UCB / UCBVI / UCRL2 deployed at the optimism-argmax are in scope without modification.*

*(A2) Multi-factor forgetting prerequisite. Within Model ( $\Sigma$ ) of Section 5.2, the bottleneck strategic tempo  $\mathcal{T}_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij}) > \rho_\Sigma / R_\Sigma$ .*

*(A3) Identifiability regime. The action mechanism is known and (C1)–(C3) of Section 5.3 hold; the agent operates in Regime A or Regime B with per-state argmax-correctness probability  $p_{\text{id}}(s)$ , with floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ .*

*(A4) Two-gap diagnostic. The agent applies the satisfaction-gap / control-regret  $2 \times 2$  to route corrective action.*

*(A5) Piecewise-stationary block structure with restart-on-change identity-tight base learner. The time axis partitions into  $B_T + 1$  stationary intervals on which the MDP is stationary and  $\pi_t^*(\cdot \mid s) =$*

340  $\delta_{a_t^*}(s)$  is fixed per state. The base learner is restarted at each block boundary; within each restarted in-  
 341 terval of length  $L$  it satisfies  $\sum_{t=1}^L \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{L}$ , where  $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid$   
 342  $s_h), Q_t(\cdot \mid s_h))]$  is the occupancy-weighted per-round coordinate. (A non-restarting carryover  
 343 variant is plausible but requires bounding cross-block occupancy shift; see Remark in Section 5.)

344 Then:

345 **(i) Per-round regret is two-sided identity-bounded.** At every visited state,  $\Delta_{\min}(1 - e^{-K_t(s)}) \leq$   
 346  $R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$ .

347 **(ii) Aggregate strategic mismatch is ultimately bounded.** Under (A2), the modeled mismatch  $\|\delta_{\Sigma}\|$  is  
 348 ultimately bounded within  $R_{\Sigma}^* = \rho_{\Sigma}/\mathcal{T}_{\Sigma}^{\text{bn,ss}}$ .

349 **(iii) The KL coordinate is computable with controlled bias.**  $-\log Q_t(a_{\text{ag},t}^*(s) \mid s)$  is computable  
 350 directly from  $Q_t$ . Per-state bias:  $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$  under  $Q_t(\cdot \mid s) \geq q_0$   
 351 at the two argmax candidates. Vanishes in Regime A.

352 **(iv) The  $2 \times 2$  cell identifies the corrective-action class.**  $(\delta_{\text{sat}}, \delta_{\text{regret}})$  routes to revise-policy / revise-  
 353 model-class-or-horizon / revise-objective.

354 **(v) Trajectory-level blockwise dynamic regret.** Under (A5),  $\mathbb{E}[\text{DynReg}(T)] :=$   
 355  $\mathbb{E}\left[\sum_{t=1}^T (V_t^{\pi_t^*} - V_t^{Q_t})\right] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) \cdot T$ .

356 The proof composes the per-round identity (i) with the simulation lemma<sup>[40–43]</sup>, a block decomposition  
 357 at optimum-change events, Cauchy–Schwarz across the  $B_T + 1$  blocks, and the bias bound from (iii).  
 358 Full proof in Appendix A; the four key lemmas are surfaced in Section 5.

### 359 4.3 Unpacking the conclusions

360 Theorem 4.1 asserts that the four components compose into a non-stationarity-aware convergence  
 361 theory with three properties that no existing strand has individually: it *handles non-stationarity*  
 362 (via (ii)/(A2)), has *explicit metric structure on policy space* (via (i)/Component 2), and is *learnable*  
 363 *on-policy* (via (iii)/(A3)). Conclusion (iv) is the connective tissue routing corrective action; conclusion  
 364 (v) is the cumulative regret rate that follows from composing the per-round metric coordinate with the  
 365 base-learner stationary rate.

366 We emphasize four things about this theorem.

367 *The per-round coordinate is exact, not an inequality.* Under the canonical scope (deterministic  $\pi^*$ ), the  
 368 per-round identity-bound (i) is  $\Delta_{\min}/V_{\max}$ -Lipschitz-equivalent to the regret. Both endpoints of the  
 369 Lipschitz envelope are *achieved* on extremal value landscapes ( $\Delta(a) = V_{\max}$  for all  $a \neq a^*$  saturates  
 370 the upper bound;  $\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$  saturates the lower). The per-round coordinate is therefore  
 371 *coordinate-optimal among bounds depending only on TV*: tightening would require value-landscape  
 372 information beyond TV. This is sharper than Pinsker’s  $\sqrt{D_{\text{KL}}/2}$  — exact instead of approximate —  
 373 and sharper than the Bretagnolle–Huber envelope  $\sqrt{1 - e^{-D_{\text{KL}}}}$  at this corner since  $x < \sqrt{x}$  on  $(0, 1)$ .

374 *The cumulative rate is in the piecewise-stationary  $B_T$  family, not the continuous-variation  $V_T$  fam-*  
 375 *ily.* The lifted rate  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  that follows under UCRL2/UCBVI base learners (Ap-  
 376 pendix D) sits in the same block-shape family as the restart-on-change variants of<sup>[1]</sup> — the route  
 377 here is through the per-round identity rather than direct optimism. It is *not directly comparable* to  
 378 the continuous-variation rate  $\tilde{O}(SA V_T^{1/3} H T^{2/3})$  of<sup>[3]</sup>: the parameter ( $B_T$  vs.  $V_T$ ), the  $T$ -scaling  
 379 ( $T^{1/2}$  vs.  $T^{2/3}$ ), the  $S, A$ -scaling ( $\sqrt{SA}$  vs.  $(SA)^{1/3}$ ), and the horizon dependence ( $N_h^2$  vs.  $N_h$ ) all  
 380 differ. Under abrupt-magnitude- $\delta$  piecewise-stationarity the two regimes coincide up to constants  
 381 ( $V_T \asymp \delta B_T$ ), and B-CS1 is sharper for fixed  $B_T$ ; outside that, Mao-style continuous-variation analysis  
 382 is sharper for fixed  $V_T$ . Generalizing the per-round-identity route to the continuous-variation regime  
 383 is open (Section 6).

384 *The bias term in (v) is the value-coordinate misidentification penalty.* The second term  $V_{\max} N_h (1 -$   
 385  $p_{\text{id}}) \cdot T$  is the per-round expected value gap between the true and agent-identified optima — bounded  
 386 by  $V_{\max}$  on the misidentification event of probability  $1 - p_{\text{id}}$ , aggregated over the  $N_h$  horizon steps. It  
 387 vanishes in Regime A ( $p_{\text{id}} \rightarrow 1$  under (C1)–(C3)); it is linear in  $T$  — vacuous as a rate but bounded by

the trivial value envelope  $V_{\max} N_h \cdot T$  — outside Regime A. In Regime B ( $p_{\text{id}} \in (0, 1)$ ), the dominant rate term is the metric one provided  $1 - p_{\text{id}}$  is small enough relative to  $\sqrt{(B_T + 1)/T}$ . In Regime C ( $p_{\text{id}} \rightarrow 0$ ) the bound saturates at the trivial envelope; algorithm flags this regime as out-of-scope for on-policy learnability (Appendix D). *Strict assumption-set weakening*: the rate (v) does *not* require Lemma 4’s two-point support condition  $Q_t(\cdot \mid s) \geq q_0$  — that condition is needed only for conclusion (iii)’s diagnostic computability of the agent’s KL coordinate, not for the cumulative-regret rate itself. *The horizon factor is the standard simulation-lemma penalty*. The  $N_h$  in conclusion (v) is the linear-in- $N_h$  per-round simulation-lemma coefficient — sharper than the  $N_h^2$  of TRPO-style worst-state bounds<sup>[44]</sup>. The lifted  $N_h^2$  in the UCRL2/UCBVI cumulative rate reflects the additional  $N_h$  in the within-block base-learner constant  $c$ .

#### 4.4 Necessity of the four components

A natural reviewer question: why exactly four? The composition is a *bundle of compatible guarantees* (each conclusion supported by its own assumptions), not a single integrated theorem in which every component is necessary for the rate. The rate (v) goes through (A5) and (i) alone; the joint properties — rate plus persistence plus learnability plus corrective-action routing — require all four.

**Without Component 2 (point-mass identity):** the per-round bound is Pinsker  $V_{\max} \sqrt{D_{\text{KL}}/2}$  or BH  $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$  — both strictly looser at this corner; Pinsker is vacuous for  $D_{\text{KL}} > 2$ . The metric layer loses exactness, and the “behavior-cloning loss against optimal trajectory” interpretation of the cumulative coordinate vanishes.

**Without Component 3 (strategic tempo):** the agent has no persistence guarantee. By the structural-class theorem on  $\mathcal{A}_{\text{decay}}$ , every count-accumulating update without forgetting eventually violates the threshold for any positive disturbance; the modeled mismatch  $\|\delta_{\Sigma}\|$  grows unboundedly. The per-round bound (i) still holds, but  $K_t(s)$  may itself drift — there is no guarantee the per-round coordinate stays bounded.

**Without Component 4 (loop-Level-2):** the bound’s RHS is computable from  $Q_t$ , but its interpretation as regret-against-true-optimum requires identifying  $a_t^*(s)$  — an interventional query under the environment’s transition kernel. In Regime C ( $\iota \approx 0$ ), the bias saturates at  $\log(1/q_0)$  per state per step, contributing  $N_h \log(1/q_0) \cdot T$  to cumulative regret — the rate (v) becomes vacuous.

**Without Component 1 (two-gap diagnostic):** the regret signal  $\delta_{\text{regret}}$  alone does not distinguish “policy revision is the right move” from “the goal is unattainable from  $M_t$ .” An agent might persist in policy revision when the model, policy class, or horizon is the actual problem (Capability-limit cell), or revise the goal when policy revision would have sufficed.

So while the rate (v) goes through (A5) and (i) alone, the *bundle* — rate plus persistence plus learnability plus corrective-action routing — requires all four. Each drop is a concrete loss of an epistemic property that the field treats as separate.

## 5 Mechanism

This section overviews the proof of Theorem 4.1 via four key lemmas chained through a block decomposition. Each lemma is stated formally; intuitions are given in plain English with the algebra deferred to the appendices. The rate (v) is the composition’s load-bearing conclusion; we sketch its derivation in four steps after surfacing the lemmas.

### 5.1 Key Lemma 1: Point-mass reverse-KL/TV identity

**Lemma 5.1** (Point-mass reverse-KL/TV identity). *For deterministic  $\pi^* = \delta_{a^*}$  (per visited state, with  $a^* = a^*(s)$  and  $Q = Q(\cdot \mid s)$ ) and any policy  $Q$ ,  $D_{\text{KL}}(\pi^* \parallel Q) = -\log Q(a^*) = -\log(1 - \text{TV}(\pi^*, Q))$ , equivalently  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \parallel Q)}$ . The identity is read in the extended real sense: when  $Q(a^*) = 0$  both sides equal 1, with the natural convention  $D_{\text{KL}} = +\infty$  and  $e^{-\infty} = 0$  — the identity holds for all policies  $Q$ , including Diracs at suboptimal actions.*

434 *Intuition.* The reverse-KL collapses under the point-mass:  $D_{\text{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)}$ ,  
 435 where the convention  $0 \log 0 = 0$  kills every  $a \neq a^*$  term, leaving  $\log \frac{1}{Q(a^*)} = -\log Q(a^*)$ . Total  
 436 variation collapses just as cleanly:  $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| = 1 - Q(a^*)$ . Combining  
 437 gives the identity. Substituting into the Bretagnolle–Huber inequality  $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$   
 438 yields  $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ , strictly on  $D_{\text{KL}} > 0$ . The identity is therefore not BH-at-equality;  
 439 it is the *exact value* at the deterministic- $\pi^*$  corner, sitting strictly below the BH envelope. Full proof  
 440 in Appendix B.

441 *Per-round regret bound.* Composed with the textbook TV-regret bound  $R(Q) \leq V_{\max} \text{TV}(\pi^*, Q)$   
 442 and the matching action-gap lower bound  $R(Q) \geq \Delta_{\min} \text{TV}(\pi^*, Q)$ , the identity yields the two-sided  
 443 characterization that gives Theorem 4.1’s conclusion (i). The bound is *coordinate-optimal among*  
 444 *bounds depending only on TV*: any tighter bound on  $R(Q)$  requires value-landscape information  
 445 beyond TV.

446 *Perturbative extension.* The deterministic- $\pi^*$  scope is not a hard wall. For  $\epsilon$ -greedy  $\pi_\epsilon^*$  with  $Q$   
 447 satisfying full-support  $Q(a) \geq q_0$ ,  $\text{TV}(\pi_\epsilon^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)} + O(\epsilon \log(1/\epsilon))$  uniformly in  
 448  $Q$ ; the proof is a triangle-inequality + expansion argument given in Appendix B.2. For softmax-  
 449 regularized  $\pi_\tau^* \propto \exp(Q_O/\tau)$ , the off-optimum mass concentrates as  $O(\exp(-\Delta_{\min}/\tau))$  and the  
 450 correction is  $O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$  — exponentially small with polynomial prefactor. The two-sided  
 451 regret bound transfers with the same correction order. Outside the perturbative regime — genuinely  
 452 high-entropy optima, tied-optimum, hard-exploration — the BH inequality becomes the relevant  
 453 general bound; Pinsker is the textbook fallback.

## 454 5.2 Key Lemma 2: Forgetting prerequisite

455 The strategic mismatch state  $\delta_\Sigma = (\delta_{ij})_{(i,j) \in E}$  in the Euclidean norm  $\|\delta_\Sigma\|^2 = \sum_{(i,j)} \delta_{ij}^2$  evolves  
 456 in discrete time as  $\mathbb{E}[\Delta \delta_{ij} \mid \delta_{ij}] = -\alpha_{ij} \delta_{ij} + w_{ij}$ ,  $|w_{ij}| \leq \rho_\Sigma / |E|^{1/2}$ , where  $\alpha_{ij} \geq 0$  is  
 457 a per-element correction rate (the strategic sector parameter),  $w_{ij}$  is a bounded disturbance, and  
 458 the disturbance budget satisfies  $\sum_{(i,j)} w_{ij}^2 \leq \rho_\Sigma^2$ . The strategic reserve  $R_\Sigma$  is the operative norm-  
 459 radius beyond which the agent loses tracking. Beta-Bernoulli edge updates instantiate this *Model* ( $\Sigma$ )  
 460 approximately; gradient-based decaying-step updates instantiate it under standard Robbins–Monro  
 461 conditions; we will refer to it as *Model* ( $\Sigma$ ) throughout.

462 **Lemma 5.2** (Multi-factor forgetting prerequisite). *Within Model* ( $\Sigma$ ) *with diagonal per-element*  
 463 *correction rates*  $\alpha_{ij} = \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij}$  *(per-element observation rate*  $\nu_{ij} \in [0, 1]$ , *regime-adjusted*  
 464 *identifiability*  $\iota_{ij} \in [0, 1]$ , *edge-update gain*  $\eta_{\text{edge},ij} > 0$ ), *under per-element exponential forgetting*  
 465 *at rate*  $\lambda_{ij} \in (0, 1)$  *giving steady-state*  $\eta_{\text{edge},ij}^{\text{ss}} \approx 1 - \lambda_{ij}$ , *ultimate boundedness of*  $\|\delta_\Sigma\|$  *within*  
 466  $R_\Sigma^* = \rho_\Sigma / \tau_\Sigma^{\text{bn,ss}}$  *is sufficient under*  $\tau_\Sigma^{\text{bn,ss}} := \min_{(i,j) \in E} (\nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})) > \rho_\Sigma / R_\Sigma$ . *The*  
 467 *threshold is sharp inside the diagonal sector model: when the inequality reverses, an adversarial*  
 468 *disturbance concentrating on the bottleneck element drives*  $\|\delta_\Sigma\|$  *beyond*  $R_\Sigma$  *within the modeled*  
 469 *dynamics.*

470 *Intuition (why bottleneck, not sum).* The Lyapunov sector product is  $\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) = \sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2$ .  
 471 Bounding this from below uniformly in  $\delta_\Sigma$  requires the largest constant  $c$  such that  $\sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2 \geq$   
 472  $c \cdot \|\delta_\Sigma\|^2$  holds for all  $\delta_\Sigma$ . The largest such  $c$  is  $\min_{(i,j)} \alpha_{ij}$ , *not*  $\sum_{(i,j)} \alpha_{ij}$  — adversarial concentration  
 473 of the disturbance on the *weakest* element saturates the bound at the bottleneck. The same effect  
 474 makes per-dimension persistence conditions sharper than scalar conditions in adaptive control<sup>[45]</sup>:  
 475 scalar tempo can overstate the available margin because the worst-case disturbance concentrates on  
 476 the slowest-correcting dimension. Ultimate boundedness then follows from Khalil’s classical sector-  
 477 Lyapunov template applied to the strategic substate. Full proof in Appendix B; the structural-class  
 478 theorem on  $\mathcal{A}_{\text{decay}}$  (gain-decay updates universally violate the threshold) is in Appendix C.6.

479 *ProST as a sector-level reduction.* The block-restart structure of ProST<sup>[8]</sup> idealizes as an *impulsive*  
 480 *system* — between scheduled update times the policy is held and the mismatch evolves under dis-  
 481 turbance alone (continuous-destabilizing in the Lyapunov  $V = \|\delta_\Sigma\|^2$ ); at each update time the  
 482 policy is revised, contracting the modeled mismatch by per-update impulse gain  $\gamma \in (0, 1]$ . The  
 483 reverse-average-dwell-time framework of<sup>[46]</sup> (Theorem 1, reverse-ADT branch) gives the threshold  
 484  $\Delta_{\max} \cdot \rho_\Sigma / R_\Sigma < -\ln(1 - \gamma)^2$  where  $\Delta_{\max}$  is the longest inter-update gap. Under uniform blocks

485  $\Delta_i = T/K$  this reduces to  $K/T > \rho_\Sigma / (2\gamma R_\Sigma)$  to leading order in  $\gamma$  — recovering ProST’s  $K/T$   
 486 form with the impulse gain made explicit. Under nonuniform schedules the impulsive form is *strictly*  
 487 *stronger*: the longest block sets the threshold, not the average. This makes a tradeoff between update  
 488 frequency  $1/\Delta_{\max}$  and per-update impulse strength  $\gamma$  visible at the sector-Lyapunov level — the same  
 489 disturbance can be tolerated either by frequent weak updates or rare strong ones.

### 490 5.3 Key Lemma 3: Closed-loop interventional access

491 **Lemma 5.3** (Loop generates interventional samples). *Let  $H_t$  denote the agent’s information state*  
 492 *at round  $t$  (model state  $M_t$ , history of observations and actions). Under (C1) positivity —  $\pi_t(a \mid$   
 493  $H_t) \geq p_{\min} > 0$  on the support of  $H_t$  for the actions of interest; (C2) sequential ignorability  
 494 —  $H_t$   $d$ -separates  $a_t$  from  $o_{t+1}$  in the mutilated graph  $G_{\bar{a}_t}$  (equivalently,  $a_t \perp\!\!\!\perp Y^{(a_t)} \mid H_t$  in  
 495 potential-outcome notation); (C3) known action mechanism — the behavior policy  $\pi_t(a \mid H_t)$  is  
 496 known: executed actions are interventions relative to the environment’s transition kernel, and the  
 497 loop generates samples from  $P(o_{t+1} \mid \text{do}(a_t = a), H_t)$  — i.e., Pearl Level-2 interventional kernels  
 498 conditional on the decision context. The substantive content for the identification claim is (C2); under  
 499 the singular-trajectory + agent-as-policy architecture of Section 3, (C1) reduces to realized-action  
 500 positivity (automatic) and (C3) to agent-policy-queryability (architectural). (C1) and (C3) are stated  
 501 explicitly for forward-compatibility with off-policy / counterfactual extensions.*

502 *Intuition.* By temporal ordering and the singular-trajectory commitment of Section 3,  $a_t$  causally  
 503 precedes  $o_{t+1}$ . Under (C1)–(C3), conditioning on the information state  $H_t$  blocks the action-selection  
 504 confounding pathway, so the conditional distribution  $P(o_{t+1} \mid a_t, H_t)$  coincides with the interven-  
 505 tional distribution  $P(o_{t+1} \mid \text{do}(a_t), H_t)$  on the support where  $\pi_t(a_t \mid H_t) > 0$ . The architecture  
 506 of the agent (Q-learning, transformer, etc.) does not enter: the loop generates conditional-Level-2  
 507 samples whenever (C1)–(C3) hold, irrespective of how the agent updates internally. Whether the  
 508 agent *exploits* this data for Level-2 reasoning is a separate algorithmic question.

509 *Three regimes of identifiability.* Three regimes<sup>[29]</sup> partition usable strength of identification: - **Regime**  
 510 **A (intervention-rich,  $\iota \approx 1$ ).** Software tests, controlled labs. do-effects identified cleanly; bound  
 511 realizable on-policy. - **Regime B (partial,  $\iota \in (0, 1)$ ).** Mixed observation-intervention. Bound  
 512 holds for the model the agent identifies; bias  $\propto 1 - \iota$ . - **Regime C (observation-only,  $\iota \approx 0$ ).**  
 513 Passive monitoring. Bound provable analytically but not realizable on-policy; observer-on-sub-agent  
 514 extensions can recover identifiability in subcases (Appendix C.8).

515 Action-generated data is Level 2 in *character*, but a clean estimate of  $P(o \mid \text{do}(a), H_t)$  requires  
 516 overcoming four typical obstacles: coverage (diverse actions tried), within-step confounding, delay  
 517 (consequences past  $t + 1$ ), and partial observability. We honor this distinction. Components 2 and  
 518 4 are jointly load-bearing: the identity of Section 5.1 gives the analytic regret-KL relationship; the  
 519 loop-Level-2 access here gives the data substrate from which  $\pi^*$  — and the bound’s RHS — can be  
 520 empirically estimated under sufficient identifiability. Neither alone makes the bound learnable.

### 521 5.4 Key Lemma 4: Bias bound under partial identifiability

522 **Lemma 5.4** (Bias bound). *Assume  $Q_t(a \mid s) \geq q_0 > 0$  for both  $a = a_{\text{ag},t}^*(s)$  (the agent’s identified*  
 523 *optimum) and  $a = \tilde{a}_t^*(s)$  (the true optimum). Let  $\hat{D}_t(s) := -\log Q_t(a_{\text{ag},t}^*(s) \mid s)$  and  $D_t^{\text{true}}(s) :=$   
 524  $-\log Q_t(\tilde{a}_t^*(s) \mid s)$ . Then  $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$ , and taking expecta-  
 525 tions:  $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq (1 - p_{\text{id}}(s)) \log(1/q_0)$ , where  $p_{\text{id}}(s) := \Pr[a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)]$ .*

526 *Intuition.* The argument is an indicator decomposition. On the matching event, the difference is  
 527 identically zero. On the misidentification event,  $\log Q_t(a_{\text{ag},t}^*(s) \mid s)$  and  $\log Q_t(\tilde{a}_t^*(s) \mid s)$  both lie  
 528 in  $[\log q_0, 0]$ , so their absolute difference is at most  $\log(1/q_0)$ . The expectation bound follows from  
 529  $\Pr[\text{misidentification event}] = 1 - p_{\text{id}}(s)$ . Bias scales *linearly* in misidentification mass — vanishes  
 530 in Regime A ( $p_{\text{id}} \rightarrow 1$ ), controlled in Regime B, saturates at  $\log(1/q_0)$  in Regime C. The argmax-  
 531 correctness probability  $p_{\text{id}}$  relates to the per-edge identifiability  $\iota$  via the policy DAG structure; in  
 532 the simple-bandit case  $p_{\text{id}} = \iota$ . Note: the support condition is *two-point* (only at the two argmax  
 533 candidates), strictly weaker than the *full-support* condition  $Q(a) \geq q_0$  for all  $a$  that the perturbative  
 534 identity (Section 5.1) requires — F.1 applies whenever the two candidates are positively-supported  
 535 regardless of  $Q_t$ ’s mass on other actions. Full proof in Appendix B.

## 5.5 Combining the lemmas: cumulative regret in four steps

We sketch the proof of conclusion (v) of Theorem 4.1 — the cumulative trajectory-level dynamic regret rate — by chaining the four key lemmas through a block decomposition.

*Step 1 — Per-round bound via the simulation lemma.* For finite-horizon  $N_h$  non-stationary MDPs with bounded reward, the standard performance-difference / simulation lemma<sup>[40–43]</sup> gives, for any two policies  $\pi_t^*$  and  $Q_t$  and initial state  $s_0$ ,  $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot | s_h), Q_t(\cdot | s_h))] = V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$ , where  $d_h^{Q_t}$  is the round- $t$  learner-induced state distribution and  $\overline{\text{TV}}_t$  is the occupancy-weighted TV defined in (A5). By Key Lemma 1 applied per state,  $\text{TV}(\pi_t^*(\cdot | s), Q_t(\cdot | s)) = 1 - e^{-K_t(s)}$  — strictly sharper than Pinsker — preserved underneath the occupancy aggregation. The horizon factor  $N_h$  here is the linear-in- $N_h$  simulation-lemma penalty, sharper than the  $N_h^2$  of TRPO-style worst-state bounds<sup>[44]</sup>.

*Step 2 — Block decomposition at piecewise-stationary segment boundaries.* Partition  $[1, T]$  at  $(P, r)$ -change events into  $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$  (the kernel-stationary-segment boundaries of Section 3). Within each block  $[\tau_i, \tau_{i+1})$  of length  $\Delta_i$  the MDP is stationary by construction. (A5) — restart-on-change base learner — gives within-block guarantee  $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$ .

*Step 3 — Cauchy–Schwarz across blocks.* Across the  $B_T + 1$  blocks,  $\sum_i \sqrt{\Delta_i} \leq \sqrt{(B_T + 1)T}$  by Cauchy–Schwarz. Multiplying by  $V_{\max} N_h$  closes the rate term:  $\sum_t \mathbb{E}[V_{\max} N_h \overline{\text{TV}}_t] \leq 2cV_{\max} N_h \sqrt{(B_T + 1)T}$ .

*Step 4 — Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap.* Steps 1–3 give the rate term against the agent’s *identified* optimum policy  $\pi_{\text{ag},t}^* := \delta_{a_{\text{ag},t}^*}$  (which is what (A5)’s base-learner guarantee delivers); the dynamic regret in (v) compares to the *true* optimum  $\tilde{\pi}_t^* := \delta_{\tilde{a}_t^*}$ . Apply the simulation lemma to these two point-mass policies: each per-step bracket reduces to  $Q_h^{\tilde{\pi}_t^*}(s_h, \tilde{a}^*) - Q_h^{\pi_{\text{ag},t}^*}(s_h, a_{\text{ag}}^*) \in [0, V_{\max}] \cdot \mathbf{1}[\text{misid at } s_h]$  (zero on the matching event by optimality of  $\tilde{a}^*$ , bounded by  $V_{\max}$  on the misidentification event). Aggregated over the  $N_h$  horizon steps with floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ , the per-round misidentification penalty is at most  $V_{\max} N_h (1 - p_{\text{id}})$ , contributing  $V_{\max} N_h (1 - p_{\text{id}}) \cdot T$  to cumulative regret — *in value coordinates* throughout, no detour through the KL coordinate. (Key Lemma 4’s KL-readout bound is the right tool for conclusion (iii)’s diagnostic computability; for the value-coordinate penalty here, indicator decomposition is direct and tighter — and the support condition  $Q_t \geq q_0$  is *not* required for this term, only for conclusion (iii).) Vanishes in Regime A.

Combining Steps 1–4 gives conclusion (v):  $\mathbb{E}[\text{DynReg}(T)] \leq 2cV_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) \cdot T$ . The Cesàro tracking corollary  $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) = O(V_{\max} N_h \sqrt{(B_T + 1)/T}) \rightarrow 0$  (when  $B_T = o(T)$  and  $p_{\text{id}} \rightarrow 1$ , i.e., Regime A) follows by dividing through by  $T$ .

**Bandit case sharpening** In the bandit special case ( $N_h = 1$ ), Thompson sampling and UCB satisfy (A5) with the *stronger logarithmic rate*  $\mathbb{E}[1 - e^{-K_t}] = O(\log t / (t \Delta_{\min}^2))$  per stationary block (Appendix D) — by Lattimore and Szepesvári 2020<sup>[27]</sup> Theorem 7.1,  $\mathbb{E}[N_a(L)] = O(\log L / \Delta_a^2)$  per suboptimal arm, aggregating to the per-round form. Cumulative regret  $O(V_{\max} (B_T + 1) \log(T / (B_T + 1)) / \Delta_{\min}^2)$  — sharper than the worst-case  $\sqrt{(B_T + 1)T}$  when  $\Delta_{\min}$  is bounded away from zero. The framework’s  $V_{\max} \cdot \text{TV}$  chain is structurally one factor of  $\Delta_{\min}$  looser than direct gap-aware UCB analysis (which gives  $O(\log T / \Delta_{\min})$  cumulative by weighting each suboptimal pull by its own  $\Delta(a)$  rather than by  $V_{\max}$ ); this is a feature of the unification — the framework’s contribution is composition, not a sharper bandit constant.

**Carryover variant of (A5)** If the base learner does *not* restart at block boundaries — e.g., a single UCRL2 instance run across the whole horizon — the within-block  $\sqrt{L}$  guarantee can fail at switch events because  $d_h^{Q_t}$  at the start of block  $i + 1$  reflects the previous block’s policy rather than a fresh exploration distribution. Bounding the cross-block occupancy shift requires either (a) detecting switches and locally re-exploring (which recovers a restart-style analysis), or (b) showing the base learner’s regret is robust to mis-calibrated state distributions. Sliding-window or adaptive-window variants<sup>[1,2]</sup> can be analyzed this way under additional structural assumptions. Theorem 4.1(v) is

585 stated for the restart case; the carryover case is plausible but requires bounding the occupancy shift,  
 586 deferred as future work.

587 Detailed full proof of (v) in Appendix A.

## 588 6 Conclusion

589 We assemble four components — the two-gap diagnostic, the point-mass reverse-KL/TV identity, the  
 590 multi-factor strategic-tempo forgetting prerequisite, and closed-loop interventional access — into a  
 591 unified non-stationary RL convergence theory. The composition delivers three properties no published  
 592 strand has individually: it *handles non-stationarity* (via the strategic-tempo persistence inequality),  
 593 has *explicit metric structure on policy space* (via the point-mass identity), and is *learnable on-policy*  
 594 (via closed-loop interventional access). The cumulative dynamic regret rate  $\tilde{O}(N_h \sqrt{(B_T + 1)T})$   
 595 matches the piecewise-stationary  $B_T$  family of<sup>[1]</sup> as a corollary; the per-round identity coordinate  
 596 is *exact* under deterministic  $\pi^*$  and strictly sharper than Pinsker and Bretagnolle–Huber at this  
 597 corner. The point-mass identity is the technical anchor — absent, to our knowledge, from the prior  
 598 RL/non-stationary corpus as an upper-bound regret coordinate.

599 We close with concluding observations and practitioner takeaways.

### 600 6.1 Concluding observations

601 **On the optimal dependencies on  $N_h$ ,  $S$ ,  $A$**  The lifted UCRL2/UCBVI rate  
 602  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  has linear-in- $N_h$  per-round penalty (the simulation lemma) plus an  
 603 additional  $N_h$  from the within-block constant  $c$ . Bernstein-type concentration in the base learner  
 604 could potentially shave one  $\sqrt{N_h}$  factor, by analogy with<sup>[43]</sup> for tabular non-stationary MDPs. We  
 605 believe the  $\sqrt{N_h}$  gap between this and the corresponding tabular non-stationary lower bound is  
 606 expected because the base-learner concentration used here is intrinsically Hoeffding-type.

607 **On the continuous-variation extension** The current rate is in the piecewise-stationary  $B_T$  family  
 608 (kernel-stationary-segment count, Section 3). Lifting to the continuous-variation  $V_T$  family of<sup>[3]</sup>  
 609 would require a different block structure — variation-aware windows with adaptive width<sup>[2]</sup> — and  
 610 a different analysis at the per-round level (continuous variation cannot be partitioned at discrete  
 611 segment-change events). It remains an interesting open question whether the per-round-identity route  
 612 can match the  $T^{2/3}$  continuous-variation rate, or whether the regimes are structurally distinct.

613 **On agent-experienced block count** The  $B_T$  of the rate (v) is an *environment-side* quantity — every  
 614 kernel-and-reward change counts. The strategic-tempo guarantee (ii) suggests an *agent-experienced*  
 615 block count  $B_T^{\text{eff}, \Sigma}$  — the number of times  $\|\delta_\Sigma\|$  exits its ultimate-bounded region — could be strictly  
 616 smaller than  $B_T$  in regimes where per-element forgetting absorbs within-segment drift. Whether (v)  
 617 sharpens to  $\sqrt{(B_T^{\text{eff}, \Sigma} + 1)T}$  requires algorithmic design coupling the base learner’s confidence-set  
 618 widening with the strategic substate  $\delta_\Sigma$ , deferred to follow-up. The result would explain why agents  
 619 in high- $V_T$  slow-drift environments often outperform  $B_T$ -pessimistic predictions: what matters is the  
 620 *agent’s effective* variation budget, not the environment’s.

621 **On joint uniqueness of the four-component composition** The metric layer is uniquely reverse-KL  
 622 up to positive scaling under chain-rule additivity (Theorem C.3 in Appendix C.10) — any theory  
 623 satisfying explicit metric structure plus chain-rule must use reverse-KL. We do *not* claim joint  
 624 uniqueness across all three properties (non-stationarity / metric / learnable). Alternative correction  
 625 architectures (non-sector-bounded revisions; non-diagonal correction) and alternative interventional-  
 626 access taxonomies exist in principle, and a joint-uniqueness theorem would require further axioms  
 627 (singular-trajectory + sector-boundedness + identifiability conditions) — open.

628 **On coupled-goal architectures** (C2)’s sequential-ignorability — that  $H_t$  d-separates  $a_t$  from  
 629  $o_{t+1}$  in the mutilated graph  $G_{\bar{a}_t}$  — presupposes architectural separation (conditional independence)  
 630 between the model state  $M_t$  (in  $H_t$ ) and the goal state. Goal-conditioned LLM policies and similar  
 631 coupled-goal architectures violate this separation: the goal influences both action selection (via the  
 632 policy) and the environment’s response to the action (via shared latent context such as user intent,

dialogue history, or task framing), so  $H_t$  does not block the goal-mediated path from  $a_t$  to  $o_{t+1}$  — (C2) fails and the loop’s data-generating process is no longer interventional in our sense. The conditional-independence relaxation results of<sup>[47]</sup> are a starting point for handling such cases. We treat this as an interesting boundary case that the current framework does not directly cover.

**On strict tightness of the strategic-tempo threshold** The bottleneck threshold  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$  is *sufficient and sharp inside the diagonal sector model* — when reversed, an adversarial disturbance concentrating on the bottleneck element drives the modeled mismatch beyond  $R_{\Sigma}$ . The theorem is silent about correction architectures *outside* the sector-Lyapunov reduction (e.g., non-diagonal correction, alternative non-sector-bounded revisions, or stabilization produced by mechanisms other than the modeled  $\alpha_{\Sigma}$ -correction). Whether the threshold extends to those architectures is a useful open question in the adaptive-control–RL bridge.

## 6.2 Practitioner takeaways

For a practitioner seeking to apply the four-component theory:

1. **The point-mass identity at the deterministic- $\pi^*$  corner is a corner-case sharpening of Bretagnolle–Huber that hasn’t been deployed in RL regret upper bounds.** Where deterministic  $\pi^*$  applies — most tabular finite-MDP regret analyses with isolated optima — the identity yields *exact* regret-vs-KL Lipschitz equivalence rather than the loose Pinsker / BH inequalities. The per-round coordinate  $1 - e^{-D_{\text{KL}}}$  is computable directly from  $Q$  and is strictly sharper than  $\sqrt{D_{\text{KL}}/2}$  at every  $D_{\text{KL}} > 0$ .
2. **The multi-factor strategic tempo lifts ProST’s tempo-as-hyperparameter framing into a structural survival inequality.** Three factors (observation rate  $\nu$ , identifiability  $\iota$ , discount rate  $1 - \lambda$ ) are independently load-bearing per element. A practitioner can use the aggregate necessary-condition  $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$  as a fail-fast pre-check and the variation-budget estimation literature<sup>[1,2,39]</sup> to estimate the environment-side  $\rho_{\Sigma}/R_{\Sigma}$  from observed reward and transition variation.
3. **Closed-loop interventional access makes regret bounds learnable on-policy under sufficient identifiability.** Regime A (intervention-rich,  $\iota \approx 1$ ) makes the bound directly realizable; Regime B (partial) carries quantified bias linear in  $1 - p_{\text{id}}$ ; Regime C (observation-only) requires observer-on-sub-agent extensions (Appendix C.8) or flags the regime as out-of-scope. The  $2 \times 2$  diagnostic routes corrective action when regret is non-zero — distinguishing “policy revision” from “model / policy class / horizon revision” from “objective revision.”

We hope this work contributes a unifying lens for the four-track non-stationary RL landscape, and that the point-mass identity is a useful tool in subsequent regret analyses where deterministic  $\pi^*$  is the canonical scope.

## References

- [1] W. C Cheung, D Simchi-Levi, and R Zhu. Reinforcement learning for non-stationary markov decision processes: the blessing of (more) optimism. *arXiv:2006.14389*, 2020.
- [2] Chen-Yu Wei and Haipeng Luo. Non-stationary reinforcement learning without prior knowledge: An optimal black-box approach, 2021.
- [3] W Mao, K Zhang, R Zhu, D Simchi-Levi, and T Başar. Near-optimal model-free rl in non-stationary episodic mdps. In *ICML*, 2021.
- [4] P Gajane, R Ortner, and P Auer. Variational regret bounds for reinforcement learning. In *UAI*, 2019.
- [5] Long-Fei Li, Peng Zhao, and Zhi-Hua Zhou. Dynamic regret of adversarial MDPs with unknown transition and linear function approximation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, 2024. doi: 10.1609/aaai.v38i12.29261.
- [6] Y Fei, Z Yang, Z Wang, and Q Xie. Dynamic regret of policy optimization in non-stationary environments. *arXiv:2007.00148*, 2020.

- [7] F. E Stradi, A Lunghi, M Castiglioni, A Marchesi, and N Gatti. Learning cmdps with non-stationary rewards and constraints. *arXiv:2405.14372*, 2024.
- [8] Hyunin Lee, Yuhao Ding, Jongmin Lee, Ming Jin, Javad Lavaei, and Somayeh Sojoudi. Tempo adaptation in non-stationary reinforcement learning. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. arXiv 2309.14989; OpenReview 7R8noSP4vL.
- [9] Hyunin Lee, Ming Jin, Javad Lavaei, and Somayeh Sojoudi. Pausing policy learning in non-stationary reinforcement learning. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, Proceedings of Machine Learning Research, 2024. arXiv 2405.16053.
- [10] A Touati and P Vincent. Efficient learning in non-stationary linear mdps (opt-wlsvi). *arXiv:2010.12870*, 2020.
- [11] Y Russac, C Vernade, and O Cappé. Weighted linear bandits for non-stationary environments. In *NeurIPS*, 2019.
- [12] A Garivier and E Moulines. On upper-confidence bound policies for non-stationary bandit problems. *arXiv:0805.3415*, 2008.
- [13] Junzhe Zhang and E Bareinboim. Mdps with unobserved confounders: a causal approach. (*Tech. report; Columbia.*), 2016.
- [14] Junzhe Zhang and Elias Bareinboim. Online reinforcement learning for mixed policy scopes. In *Advances in Neural Information Processing Systems*, 2022.
- [15] Y Lu, A Meisami, and A Tewari. Causal mdps: learning good interventions efficiently. *arXiv:2102.07663*, 2021.
- [16] Y Lu, A Meisami, and A Tewari. Efficient rl with prior causal knowledge. *CLEaR*, 2022.
- [17] L Wang, Z Yang, and Z Wang. Provably efficient causal rl with confounded observational data (dovi). In *NeurIPS*. *arXiv:2006.12311 (2020)*, 2021.
- [18] Junzhe Zhang. Designing optimal dynamic treatment regimes: a causal rl approach. In *ICML*, 2020.
- [19] D Russo and B Van Roy. An information-theoretic analysis of thompson sampling. *J. Mach. Learn. Res.*, 2014.
- [20] Daniel Russo and Benjamin Van Roy. Learning to optimize via information-directed sampling. *Operations Research*, 2014.
- [21] X Lu and B Van Roy. Information-theoretic confidence bounds for reinforcement learning. In *NeurIPS*, 2019.
- [22] Seungki Min and Daniel Russo. An information-theoretic analysis of nonstationary bandit learning. In *International Conference on Machine Learning (ICML 2023)*, 2023.
- [23] T Lattimore and A György. Mirror descent and the information ratio. In *COLT. (PMLR vol 134)*; *arXiv:2009.12228 (2020)*, 2021.
- [24] Sham Kakade, Akshay Krishnamurthy, Kendall Lowrey, Motoya Ohnishi, and Wen Sun. Information theoretic regret bounds for online nonlinear control. In *Advances in Neural Information Processing Systems*, 2020. arXiv:2006.12466.
- [25] Jean Bretagnolle and Catherine Huber. Estimation des densités: Risque minimax. In C. Del-lacherie, P. A. Meyer, and M. Weil, editors, *Séminaire de Probabilités XII*, volume 649 of *Lecture Notes in Mathematics*, pages 342–363. Springer, 1978. doi: 10.1007/BFb0064609.
- [26] Alexandre B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer Series in Statistics. Springer, New York, 2009. doi: 10.1007/b13794.
- [27] T Lattimore and C Szepesvári. *Bandit Algorithms*. Cambridge., 2020.

- [28] Hassan K. Khalil. *Nonlinear Systems*. Prentice-Hall, Upper Saddle River, NJ, 3rd edition, 2002.
- [29] Elias Bareinboim, Juan D. Correa, Duligur Ibeling, and Thomas Icard. On Pearl’s hierarchy and the foundations of causal inference. In Hector Geffner, Rina Dechter, and Joseph Y. Halpern, editors, *Probabilistic and Causal Inference: The Works of Judea Pearl*, pages 507–556. ACM Books, 2022. doi: 10.1145/3501714.3501743.
- [30] Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. Active inference: a process theory. *Neural Computation*, 29(1):1–49, 2017.
- [31] T Parr, G Pezzulo, and K. J Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press., 2022.
- [32] Wang Chi Cheung, David Simchi-Levi, and Ruihao Zhu. Reinforcement learning for non-stationary Markov decision processes: The blessing of (more) optimism. *Management Science*, 68(3):1697–1716, 2022. doi: 10.1287/mnsc.2021.4123. ArXiv 2020.
- [33] Y Yang, Z Zheng, M Tomizuka, C Liu, and S Li. The feasibility theory of constrained reinforcement learning: A tutorial study. *arXiv:2404.10064*, 2024.
- [34] Oliver Schulte and Pascal Poupart. When should reinforcement learning use causal reasoning? *Transactions on Machine Learning Research*, 2025. Published 2025-05-09; OpenReview D1PPuk8ZBI; Survey Certification.
- [35] J Pearl. *Causality: Models, Reasoning, and Inference*. (2nd ed.). Cambridge., 2009.
- [36] C Canonne. A short note on an inequality between kl and tv. *arXiv:2202.07198*, 2022.
- [37] H Hajiabolhassan and R Ortner. Online regret bounds for satisficing in markov decision processes. *Mathematics of Operations Research*, 2025.
- [38] Y. Zhang, R Zhu, and Q Xie. On the peril of (even a little) non-stationarity in satisficing regret minimization. (*contemporaneous; March 2026*), 2026.
- [39] A Gerogiannis, Huang, Y.-H, and V Veeravalli. Darling: Detection augmented reinforcement learning with non-stationary guarantees. (*contemporaneous; April 2026*), 2026.
- [40] S Kakade and J Langford. Approximately optimal approximate reinforcement learning (cpi / performance-difference lemma). In *ICML*, 2002.
- [41] R Munos. Error bounds for approximate policy iteration. In *ICML*, 2003.
- [42] S Ross and J. A Bagnell. Efficient reductions for imitation learning (dagger). In *AISTATS*, 2010.
- [43] M. G Azar, I Osband, and R Munos. Minimax regret bounds for reinforcement learning (ucbvi). In *ICML*. *arXiv:1703.05449*, 2017.
- [44] J Schulman, S Levine, P Moritz, M. I Jordan, and P Abbeel. Trust region policy optimization (trpo). In *ICML*. *arXiv:1502.05477*, 2015.
- [45] Brian D. O. Anderson. Adaptive systems, lack of persistency of excitation and bursting phenomena. *Automatica*, 21(3):247–258, 1985. doi: 10.1016/0005-1098(85)90058-5.
- [46] J. P Hespanha, D Liberzon, and A. R Teel. Lyapunov conditions for input-to-state stability of impulsive systems. *Automatica*. 44(11):2735–2744, 2008.
- [47] Jelle Bruineberg, Krzysztof Dolega, Joe Dewhurst, and Manuel Baltieri. The Emperor’s new Markov blankets. *Behavioral and Brain Sciences*, 45:e183, 2022. doi: 10.1017/S0140525X21002351.
- [48] S Levine. Reinforcement learning and control as probabilistic inference. *arXiv:1805.00909*, 2018.
- [49] N Wiener. *Cybernetics*. MIT Press., 1948.

[50] R Conant and W. R Ashby. Every good regulator of a system must be a model of that system. *Int. J. Systems Sci.*, 1970.

[51] J Aczél and Z Daróczy. *On Measures of Information and Their Characterizations*. Academic Press., 1975.

[52] P Auer, T Jaksch, and R Ortner. Near-optimal regret bounds for reinforcement learning (ucrl2). *J. Mach. Learn. Res.* 11:1563–1600, 2010.

## A Proof of the Composition Theorem

This appendix gives the full proof of Theorem 4.1. The proof composes the four key lemmas of Section 5 (full proofs of those in Appendix B) with a block decomposition at optimum-change events plus Cauchy–Schwarz across blocks.

**Notation** Throughout this section:  $K_t(s) := D_{\text{KL}}(\delta_{a_t^*(s)} \parallel Q_t(\cdot \mid s))$  is the per-state reverse-KL coordinate;  $d_h^{Q_t}(\cdot \mid s_0)$  is the round- $t$  learner-induced state distribution at horizon step  $h$  starting from  $s_0$ ;  $\overline{\text{TV}}_t := \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [\text{TV}(\pi_t^*(\cdot \mid s_h), Q_t(\cdot \mid s_h))]$  is the occupancy-weighted per-round coordinate of (A5);  $B_T = |\{t : (P_t, r_t) \neq (P_{t-1}, r_{t-1})\}|$  counts piecewise-stationary segment boundaries (per Section 3); the partition is  $0 = \tau_0 < \tau_1 < \dots < \tau_{B_T} \leq \tau_{B_T+1} = T$  with block  $i$  length  $\Delta_i := \tau_{i+1} - \tau_i$ .

### A.1 Proof of (i) — Per-round identity-bounded regret

For deterministic  $\pi_t^* = \delta_{a_t^*(s)}$  and any  $Q_t(\cdot \mid s)$ , Key Lemma 1 (Lemma 5.1, full proof in Appendix B.1) gives the exact identity  $\text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) = 1 - e^{-K_t(s)}$  in the extended real sense (when  $Q_t(a_t^*(s) \mid s) = 0$ ,  $K_t(s) = +\infty$  and both sides equal 1, recovering the trivial bound). Composed with the per-state TV-regret bounds (full proofs in Appendix B.1 and Appendix C.4):  $\Delta_{\min} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s)) \leq R(Q_t \mid s) \leq V_{\max} \cdot \text{TV}(\delta_{a_t^*(s)}, Q_t(\cdot \mid s))$ , substituting the identity yields conclusion (i):  $\Delta_{\min}(1 - e^{-K_t(s)}) \leq R(Q_t \mid s) \leq V_{\max}(1 - e^{-K_t(s)})$ .  $\square$

### A.2 Proof of (ii) — Aggregate mismatch ultimate boundedness

Under (A2), Key Lemma 2 (Lemma 5.2, full proof in Appendix B.3) applied to the diagonal sector model with  $\alpha_{ij}^{\text{ss}} = \nu_{ij} \cdot \iota_{ij} \cdot (1 - \lambda_{ij})$  gives ultimate boundedness of  $\|\delta_\Sigma\|$  within  $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$ .  $\square$

### A.3 Proof of (iii) — KL coordinate bias bound

Key Lemma 4 (Lemma 5.4, full proof in Appendix B.6) gives the per-state bias bound directly, under (A3)’s identifiability assumption (which gives  $p_{\text{id}}(s)$  for each  $s$ ) and the two-point support condition  $Q_t(\cdot \mid s) \geq q_0$  at  $a_{\text{ag},t}^*(s)$  and  $\tilde{a}_t^*(s)$ .  $\square$

### A.4 Proof of (iv) — $2 \times 2$ corrective-action routing

(A4) states the agent applies the satisfaction-gap / control-regret  $2 \times 2$  (Section 4.1 Component 1). Conclusion (iv) is a direct restatement of the  $2 \times 2$  cell partition. The decomposition is convention-dependent (the values of  $\delta_{\text{sat}}$  and  $\delta_{\text{regret}}$  depend on the continuation convention; see Appendix C.1 for monotonicity across C1/C2/C3); the  $2 \times 2$  structure is preserved across all three conventions.  $\square$

### A.5 Proof of (v) — Trajectory-level cumulative regret

We chain Key Lemmas 1 and 4 with the simulation lemma and a Cauchy–Schwarz block argument. The proof is in four steps; the high-level sketch is in Section 5.5.

*Step 1: Simulation / performance-difference lemma.* For any two policies  $\pi$  and  $\pi'$ , any state  $s$ , and finite-horizon non-discounted MDP with horizon  $N_h$ , the standard performance-difference lemma<sup>[40–43]</sup> gives  $V_t^{\pi'}(s) - V_t^\pi(s) =$

810  $\mathbb{E} \left[ \sum_{h=0}^{N_h-1} \left( \mathbb{E}_{a' \sim \pi'(\cdot | s_h)} [Q_h^{\pi'}(s_h, a')] - Q_h^{\pi'}(s_h, a_h) \right) \mid s_0 = s, a_h \sim \pi \right]$ . Each per-step bracket  
811 is bounded by value range times TV:  $|\mathbb{E}_{a' \sim \pi'} [Q_h^{\pi'}] - \mathbb{E}_{a \sim \pi} [Q_h^{\pi'}]| \leq V_{\max} \cdot \text{TV}(\pi'(\cdot | s_h), \pi(\cdot | s_h))$ . Applied at round  $t$  with  $\pi' = \pi_t^*$ ,  $\pi = Q_t$ :  $V_t^{\pi_t^*}(s_0) - V_t^{Q_t}(s_0) \leq$   
812  $V_{\max} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}(\cdot | s_0)} [\text{TV}(\pi_t^*(\cdot | s_h), Q_t(\cdot | s_h))] = V_{\max} \cdot N_h \cdot \overline{\text{TV}}_t$ . The horizon factor  $N_h$   
813 here is the *linear-in- $N_h$*  simulation-lemma penalty — sharper than the  $N_h^2$  of TRPO-style worst-state  
814 bounds<sup>[44]</sup>, which bound max-state TV rather than occupancy-weighted TV.

816 *Step 2: Per-state TV identity along the trajectory.* By the deterministic-per-state optimum scope of  
817 Theorem 4.1 ( $\pi_t^*(\cdot | s) = \delta_{a_t^*}(s)$ ), at every visited state Key Lemma 1 gives  $\text{TV}(\pi_t^*(\cdot | s), Q_t(\cdot | s)) = 1 - Q_t(a_t^*(s) | s) = 1 - e^{-K_t(s)}$ . Summing over the trajectory and taking expectations:  
818  $\overline{\text{TV}}_t = \frac{1}{N_h} \sum_{h=0}^{N_h-1} \mathbb{E}_{s_h \sim d_h^{Q_t}} [1 - e^{-K_t(s_h)}]$ . The per-state KL/TV identity is preserved underneath  
819 the occupancy aggregation; the per-round coordinate is sharper than Pinsker/BH at every visited state.

821 *Step 3: Block-sum + Cauchy-Schwarz across blocks.* Within each block  $[\tau_i, \tau_{i+1})$  of length  
822  $\Delta_i$ , (A5)’s restart-on-change identity-tight base learner gives  $\sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c\sqrt{\Delta_i}$ . Sum-  
823 ming over the  $B_T + 1$  blocks:  $\sum_{t=1}^T \mathbb{E}[\overline{\text{TV}}_t] = \sum_{i=0}^{B_T} \sum_{t \in \text{block}_i} \mathbb{E}[\overline{\text{TV}}_t] \leq 2c \sum_{i=0}^{B_T} \sqrt{\Delta_i}$ .  
824 Cauchy-Schwarz gives  $\sum_{i=0}^{B_T} \sqrt{\Delta_i} \leq \sqrt{(B_T + 1) \cdot \sum_i \Delta_i} = \sqrt{(B_T + 1)T}$ . Multiplying by  
825  $V_{\max} N_h$ :  $\sum_{t=1}^T \mathbb{E}[V_{\max} N_h \overline{\text{TV}}_t] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T}$ . This is the rate term.

826 *Step 4: Misidentification penalty via simulation lemma at the identified-vs-true-optimum gap.* Steps  
827 1–3 implicitly compared the agent’s policy  $Q_t$  to the *agent’s identified-optimum policy*  $\pi_{\text{ag},t}^*(\cdot | s)$   
828  $:= \delta_{a_{\text{ag},t}^*}(s)$  (since (A5)’s base-learner guarantee is what the agent achieves on its identified  
829 objective). The trajectory-level dynamic regret in (v) compares to the *true* optimum  $\tilde{\pi}_t^*(\cdot | s) := \delta_{\tilde{a}_t^*}(s)$ .

830 Decompose:  $V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{Q_t}(s_0) = [V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{\pi_{\text{ag},t}^*}(s_0)] + [V_t^{\pi_{\text{ag},t}^*}(s_0) - V_t^{Q_t}(s_0)]$ . Steps  
831 1–3 bound the second bracket. The first bracket is the misidentification penalty. Apply the simulation  
832 lemma to the two point-mass policies  $(\tilde{\pi}_t^*, \pi_{\text{ag},t}^*)$ : each per-step bracket reduces to  $Q_h^{\tilde{\pi}_t^*}(s_h, \tilde{a}_t^*(s_h)) -$   
833  $Q_h^{\pi_{\text{ag},t}^*}(s_h, a_{\text{ag},t}^*(s_h)) \in [0, V_{\max}] \cdot \mathbf{1}[\tilde{a}_t^*(s_h) \neq a_{\text{ag},t}^*(s_h)]$ . Lower bound by optimality of  $\tilde{a}^*$ ; upper  
834 bound by the cumulative-Q range with the indicator on the misidentification event. Taking expectations,  
835 with the per-state floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ :  $\mathbb{E}[V_t^{\tilde{\pi}_t^*}(s_0) - V_t^{\pi_{\text{ag},t}^*}(s_0)] \leq V_{\max} \cdot N_h \cdot (1 - p_{\text{id}})$ . Summing  
836 over  $T$  rounds:  $V_{\max} \cdot N_h \cdot (1 - p_{\text{id}}) \cdot T$ . This is the bias term, in *value coordinates* throughout — no  
837 detour through the KL coordinate. Key Lemma 4’s KL-readout bound is the right tool for conclusion  
838 (iii)’s diagnostic computability; for the value-coordinate misidentification penalty here, the indicator  
839 decomposition is direct and tighter (the support condition  $Q_t \geq q_0$  at the two argmax candidates is  
840 *not* required for this term — only conclusion (iii) needs it).

841 *Combining.* Adding Steps 1–4:  $\mathbb{E}[\text{DynReg}(T)] \leq 2c V_{\max} N_h \sqrt{(B_T + 1)T} + V_{\max} N_h (1 - p_{\text{id}}) T$ ,  
842 which is conclusion (v).  $\square$

843 **Cesàro tracking corollary** Dividing through by  $T$ :  $\frac{1}{T} \sum_t (V_t^{\tilde{\pi}_t^*} - V_t^{Q_t}) \leq$   
844  $2c V_{\max} N_h \sqrt{(B_T + 1)/T} + V_{\max} N_h (1 - p_{\text{id}})$ . As  $T \rightarrow \infty$  with  $B_T = o(T)$  and  $p_{\text{id}} \rightarrow 1$   
845 (Regime A), the right side tends to zero — the agent’s average value over the trajectory tracks the  
846 moving optimum’s average value. In Regime B the second term is constant; tracking is only up to a  
847  $V_{\max}(1 - p_{\text{id}})$  residual.  $\square$

848 **Bandit case sharpening (Remark)** In the bandit special case ( $N_h = 1$ , so  $\overline{\text{TV}}_t = \mathbb{E}[1 - e^{-K_t}]$ ),  
849 Thompson sampling and UCB (Appendix D) satisfy (A5) with the *stronger logarithmic* per-block rate.  
850 By Lattimore and Szepesvári 2020<sup>[27]</sup> Theorem 7.1 (or [? auer-cesa-bianchi-fischer-2002-finitetime]),  
851  $\mathbb{E}[N_a(L)] = O(\log L / \Delta_a^2)$  per suboptimal arm  $a$ ; aggregating over arms gives  $\mathbb{E}[1 - e^{-K_t}] =$   
852  $\mathbb{E}[1 - Q_t(a^*)] = O(\log t / (t \Delta_{\min}^2))$  per stationary block. Substituting and summing gives cumulative  
853 regret  $O(V_{\max}(B_T + 1) \log(T/(B_T + 1)) / \Delta_{\min}^2)$  — sharper than the worst-case  $\sqrt{(B_T + 1)T}$  when  
854  $\Delta_{\min}$  is bounded away from zero. The framework’s  $V_{\max} \cdot \text{TV}$  chain is structurally one factor of  $\Delta_{\min}$   
855 looser than direct gap-aware UCB analysis (which gives  $O(\log T / \Delta_{\min})$  cumulative regret per block  
856 by weighting each suboptimal pull by  $\Delta(a)$  rather than  $V_{\max}$ ); this is a feature of the unification, not a  
857 bug. For  $N_h > 1$  MDPs, UCRL2 and UCBVI provide the natural occupancy-weighted instantiation

(Appendix D) with  $c$  of order  $\tilde{O}(\sqrt{N_h SA})$ , lifting cleanly to  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  — same piecewise-stationary  $B_T$  family as restart-on-change variants of<sup>[1]</sup>.

## B Proofs of the Key Lemmas

This appendix gives full proofs of the four key lemmas surfaced in Section 5, plus the perturbative extension and the ProST sector-level reduction.

### B.1 Proof of Key Lemma 1 (Point-mass reverse-KL/TV identity)

For deterministic  $\pi^* = \delta_{a^*}$  and any policy  $Q$  (extending  $D_{\text{KL}} = +\infty$  when  $Q(a^*) = 0$ , which gives  $1 - e^{-D_{\text{KL}}} = 1 = \text{TV}$  trivially):

The reverse-KL collapses under the point-mass:  $D_{\text{KL}}(\delta_{a^*} \| Q) = \sum_a \delta_{a^*}(a) \log \frac{\delta_{a^*}(a)}{Q(a)} = \log \frac{1}{Q(a^*)} = -\log Q(a^*)$ , where the convention  $0 \log 0 = 0$  disposes of the  $a \neq a^*$  terms. Combined with the textbook total-variation calculation  $\text{TV}(\delta_{a^*}, Q) = \frac{1}{2} \sum_a |\delta_{a^*}(a) - Q(a)| = \frac{1}{2} (|1 - Q(a^*)| + \sum_{a \neq a^*} Q(a)) = \frac{1}{2} ((1 - Q(a^*)) + (1 - Q(a^*))) = 1 - Q(a^*)$ , we have  $-\log Q(a^*) = -\log(1 - \text{TV})$ . Solving for TV gives  $\text{TV} = 1 - e^{-D_{\text{KL}}}$ .  $\square$

**Strict-improvement substitution** Substituting the identity into the Bretagnolle–Huber inequality  $\text{TV}(P, Q) \leq \sqrt{1 - e^{-D_{\text{KL}}(P \| Q)}}$  at  $P = \delta_{a^*}$  yields  $1 - e^{-D_{\text{KL}}} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$ , which holds *strictly* on  $D_{\text{KL}} > 0$  since  $x < \sqrt{x}$  on  $(0, 1)$ . The identity therefore supplies the *exact* TV value at the deterministic- $\pi^*$  corner and lies strictly below the BH envelope there.

**Two-sided regret bound** (stated as Component 2’s two-sided characterization in Section 4.1 and as Key Lemma 1’s *Per-round regret bound* paragraph in Section 5.1). From the regret expression  $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a)$  and  $\Delta(a) \leq V_{\max}$ :  $R(Q) \leq V_{\max} \sum_{a \neq a^*} Q(a) = V_{\max} (1 - Q(a^*)) = V_{\max} \text{TV}(\delta_{a^*}, Q) = V_{\max} (1 - e^{-D_{\text{KL}}})$ . The matching action-gap lower bound (full statement in Appendix C.4) gives  $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \text{TV}(\delta_{a^*}, Q) = \Delta_{\min} (1 - e^{-D_{\text{KL}}})$ . Combining gives the two-sided characterization. The bound is *Lipschitz-equivalent* with constants  $\Delta_{\min}/V_{\max}$  and 1, both achieved on extremal value landscapes — therefore *coordinate-optimal among bounds depending only on TV*.  $\square$

### B.2 Proof of the perturbative extension — $\epsilon$ -stochastic and softmax-regularized

We establish the perturbative identity via a uniform perturbation argument rather than an alignment-specific calculation.

**$\epsilon$ -greedy stochastic optimum** Let  $\pi_\epsilon^*(a^*) = 1 - \epsilon$  and  $\pi_\epsilon^*(a) = \epsilon/(|\mathcal{A}| - 1)$  for  $a \neq a^*$ , and assume the full-support lower bound  $Q(a) \geq q_0 > 0$  for all  $a \in \mathcal{A}$ . (Full support is required since  $\pi_\epsilon^*$  has positive mass on every action; without  $Q(a) > 0$  on the full support,  $D_{\text{KL}}(\pi_\epsilon^* \| Q) = +\infty$ .)

*Step 1 — TV is Lipschitz under the point-mass perturbation.*  $\text{TV}(\pi_\epsilon^*, \delta_{a^*}) = \epsilon$ . By the triangle inequality for total variation,  $|\text{TV}(\pi_\epsilon^*, Q) - \text{TV}(\delta_{a^*}, Q)| \leq \text{TV}(\pi_\epsilon^*, \delta_{a^*}) = \epsilon$ , uniformly in  $Q$  — no alignment hypothesis needed.

*Step 2 — Reverse-KL admits a uniform expansion.* Direct computation:

$$\begin{aligned} D_{\text{KL}}(\pi_\epsilon^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q) &= (1 - \epsilon) \log \frac{1 - \epsilon}{Q(a^*)} + \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log \frac{\epsilon/(|\mathcal{A}| - 1)}{Q(a)} - (-\log Q(a^*)) \\ &= (1 - \epsilon) \log(1 - \epsilon) + \epsilon \log Q(a^*) + \epsilon \log \frac{\epsilon}{|\mathcal{A}| - 1} - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a), \end{aligned}$$

where the second equality collects the  $\log Q(a^*)$  contributions via  $-(1 - \epsilon) \log Q(a^*) + \log Q(a^*) = \epsilon \log Q(a^*)$  and splits the constant log term off the conditional sum. Expanding  $(1 - \epsilon) \log(1 - \epsilon) = -\epsilon + O(\epsilon^2)$  and  $\epsilon \log(\epsilon/(|\mathcal{A}| - 1)) = \epsilon \log \epsilon - \epsilon \log(|\mathcal{A}| - 1)$ , the leading-order behavior is  $D_{\text{KL}}(\pi_\epsilon^* \| Q) = D_{\text{KL}}(\delta_{a^*} \| Q) + \epsilon \log \epsilon - \epsilon (\log(|\mathcal{A}| - 1) + 1) + \epsilon \log Q(a^*) - \frac{\epsilon}{|\mathcal{A}| - 1} \sum_{a \neq a^*} \log Q(a) + O(\epsilon^2)$ . Under  $Q(a) \geq q_0$ , each

899  $|\log Q(a)| \leq \log(1/q_0)$  is uniformly bounded; the dominant correction is the  $\epsilon \log \epsilon$  leading  
900 term, giving  $|D_{\text{KL}}(\pi_\epsilon^* \| Q) - D_{\text{KL}}(\delta_{a^*} \| Q)| = O(\epsilon \log(1/\epsilon) + \epsilon \log(1/q_0)) = O(\epsilon \log(1/\epsilon))$   
901 absorbing  $\epsilon \log(1/q_0)$  into the leading  $\epsilon \log(1/\epsilon)$  term as  $\epsilon \rightarrow 0$  (constants depend on  $q_0$  and  $|\mathcal{A}|$ ).

902 *Step 3 — Map through  $1 - e^{-x}$  via Lipschitzness.* The map  $x \mapsto 1 - e^{-x}$  is 1-Lipschitz on  $[0, \infty)$ ,  
903 so  $|(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) - (1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)})| = O(\epsilon \log(1/\epsilon))$ . Combining Steps 1 and 3 with  
904 the unperturbed identity  $\text{TV}(\delta_{a^*}, Q) = 1 - e^{-D_{\text{KL}}(\delta_{a^*} \| Q)}$ :  $\text{TV}(\pi_\epsilon^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)} +$   
905  $O(\epsilon \log(1/\epsilon))$  uniformly in  $Q$  over the class  $\{Q : Q(a) \geq q_0\}$ , with constants depending on  $q_0$  and  
906  $|\mathcal{A}|$ . This establishes the  $\epsilon$ -greedy form of the perturbative identity.  $\square$

907 **Softmax-regularized optimum** Let  $\pi_\tau^*(a) \propto \exp(Q_O(a)/\tau)$ . Under isolated optimum with  
908 action gap  $\Delta_{\min}$ , the off-optimum mass concentrates exponentially in  $1/\tau$ :  $1 - \pi_\tau^*(a^*) =$   
909  $\frac{\sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)}{1 + \sum_{a \neq a^*} \exp(-(Q_O(a^*) - Q_O(a))/\tau)} \leq (|\mathcal{A}| - 1) \exp(-\Delta_{\min}/\tau)$ . Treating  $\pi_\tau^*$  as  $\pi_\epsilon^*$  with ef-  
910 fective  $\epsilon = O(\exp(-\Delta_{\min}/\tau))$ , the  $\epsilon$ -greedy expansion above applies with  $\epsilon \log(1/\epsilon) =$   
911  $O((\Delta_{\min}/\tau) \exp(-\Delta_{\min}/\tau)) = O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$ , exponentially small with a polynomial  
912 prefactor (equivalently,  $O(\exp(-c\Delta_{\min}/\tau))$  for any fixed  $c \in (0, 1)$  since the exponential dominates  
913 the polynomial).  $\square$

914 **Two-sided regret bound under perturbation** Composing with the TV-regret bounds:  $\Delta_{\min}(1 -$   
915  $e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) - O(\epsilon \log(1/\epsilon)) \leq R(Q) \leq V_{\max}(1 - e^{-D_{\text{KL}}(\pi_\epsilon^* \| Q)}) + O(\epsilon \log(1/\epsilon))$ .

### 916 B.3 Proof of Key Lemma 2 (Multi-factor forgetting prerequisite)

917 We work in the continuous-time idealization of Model  $(\Sigma)$ :  $\dot{\delta}_\Sigma = -\mathbf{F}(\delta_\Sigma) + \mathbf{w}$  with  $\mathbf{F}(\delta_\Sigma) :=$   
918  $(\alpha_{ij}^{\text{ss}} \delta_{ij})_{(i,j) \in E}$  the sector correction and  $\mathbf{w}$  the deterministic disturbance with  $\|\mathbf{w}\| \leq \rho_\Sigma$ . The  
919 discrete-time form recovers the same scaling up to absorbable constants via an AM-GM expansion of  
920  $\|\delta_\Sigma - \mathbf{F} + \mathbf{w}\|^2$  (sketched below).

921 For the Lyapunov function  $V(\delta_\Sigma) := \|\delta_\Sigma\|^2$ ,  $\dot{V} = 2\delta_\Sigma^\top \dot{\delta}_\Sigma = -2\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) + 2\delta_\Sigma^\top \mathbf{w}$ . The  
922 first term is the *sector product*, bounded below by the bottleneck:  $\delta_\Sigma^\top \mathbf{F}(\delta_\Sigma) = \sum_{(i,j)} \alpha_{ij}^{\text{ss}} \delta_{ij}^2 \geq$   
923  $\min_{(i,j)} \alpha_{ij}^{\text{ss}} \cdot \|\delta_\Sigma\|^2 = \mathcal{T}_\Sigma^{\text{bn,ss}} \cdot \|\delta_\Sigma\|^2$ , with the bound tight: setting  $\delta_\Sigma$  proportional to  $\mathbf{e}_{(i^*, j^*)}$  where  
924  $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$  saturates it. The cross-term is bounded by Cauchy-Schwarz:  $|2\delta_\Sigma^\top \mathbf{w}| \leq$   
925  $2\|\delta_\Sigma\| \cdot \|\mathbf{w}\| \leq 2\rho_\Sigma \|\delta_\Sigma\|$ . Combining:  $\dot{V} \leq -2\mathcal{T}_\Sigma^{\text{bn,ss}} \|\delta_\Sigma\|^2 + 2\rho_\Sigma \|\delta_\Sigma\| = -2\|\delta_\Sigma\| \cdot$   
926  $(\mathcal{T}_\Sigma^{\text{bn,ss}} \|\delta_\Sigma\| - \rho_\Sigma)$ . The drift is strictly negative whenever  $\|\delta_\Sigma\| > \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$ ; the trajectory is  
927 ultimately bounded by  $R_\Sigma^* = \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$ . So whenever  $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma / R_\Sigma$ , we have  $R_\Sigma^* \leq R_\Sigma$  and  
928 the modeled mismatch is ultimately bounded within the strategic reserve. This is the standard  
929 Khalil sector-Lyapunov ultimate-boundedness setup with the cross-term retained<sup>[28]</sup> (Section 9.2;  
930 comparison-principle form).  $\square$

931 *Discrete-time form.* For the discrete update  $\delta_\Sigma^+ = \delta_\Sigma - \mathbf{F}(\delta_\Sigma) + \mathbf{w}$  with  $\alpha_{ij}^{\text{ss}} \in [0, 1]$  and AM-GM  
932 with parameter  $\eta = \mathcal{T}_\Sigma^{\text{bn,ss}} / (2(1 - \mathcal{T}_\Sigma^{\text{bn,ss}}))$ , the steady-state bound is  $\|\delta_\Sigma^*\| \leq 2\rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$  — same  
933 scaling, factor-of-2 absorbable into the threshold constant.

934 *Mean-square corollary.* If instead  $\mathbf{w}$  is interpreted as zero-mean stochastic disturbance with  $\mathbb{E}[\|\mathbf{w}\|^2] \leq$   
935  $\rho_\Sigma^2$ , the cross-term  $\mathbb{E}[2\delta_\Sigma^\top \mathbf{w} \mid \delta_\Sigma] = 0$  vanishes in expectation, and the drift inequality gives mean-  
936 square ultimate boundedness  $\sqrt{\mathbb{E}[V^*]} \leq \rho_\Sigma / \sqrt{2\mathcal{T}_\Sigma^{\text{bn,ss}}}$  under threshold  $\mathcal{T}_\Sigma^{\text{bn,ss}} > \rho_\Sigma^2 / (2R_\Sigma^2)$ . The  
937 deterministic-disturbance form above is the operative one for Theorem 4.1(A2) since non-stationarity-  
938 induced drift is structurally adversarial-aligned (consistently moving mismatch in one direction) rather  
939 than zero-mean.

940 **Sharpness** When the inequality reverses ( $\mathcal{T}_\Sigma^{\text{bn,ss}} \leq \rho_\Sigma / R_\Sigma$ ), the adversarial disturbance  $\mathbf{w}^*(t) =$   
941  $\rho_\Sigma \delta_\Sigma(t) / \|\delta_\Sigma(t)\|$  saturates Cauchy-Schwarz and gives  $d\|\delta_\Sigma\|/dt \geq -\mathcal{T}_\Sigma^{\text{bn,ss}} \|\delta_\Sigma\| + \rho_\Sigma$ . By the  
942 comparison principle,  $\|\delta_\Sigma(t)\|$  exceeds any  $R < \rho_\Sigma / \mathcal{T}_\Sigma^{\text{bn,ss}}$  in finite time. The bottleneck-element  
943 refinement: pick  $\mathbf{w}$  and  $\delta_\Sigma$  both supported on  $(i^*, j^*) \in \arg \min_{(i,j)} \alpha_{ij}^{\text{ss}}$  with  $|w_{i^* j^*}| = \rho_\Sigma$  —  
944 saturates the sector inequality on the contracting side and aligns with  $\mathbf{w}$  on the disturbance side

945 simultaneously. The threshold is *sharp inside the diagonal sector model*. The theorem is silent about  
 946 non-diagonal correction architectures or stabilization mechanisms outside Model ( $\Sigma$ ).  $\square$

#### 947 B.4 Proof of the impulsive ProST sector-level reduction

948 Within Model ( $\Sigma$ ), idealize ProST<sup>[8]</sup> as an impulsive system: between scheduled update times  
 949  $\{t_1, \dots, t_K\} \subset [0, T]$  the agent’s policy is held fixed and the strategic mismatch evolves under  
 950 disturbance budget  $\rho_\Sigma$  alone (continuous-destabilizing in the Lyapunov  $V = \|\delta_\Sigma\|^2$ ); at each update  
 951 time  $t_i$  the policy is revised, contracting the modeled mismatch by per-update impulse gain  $\gamma \in (0, 1]$   
 952 so  $\|\delta_\Sigma(t_i^+)\| \leq (1 - \gamma)\|\delta_\Sigma(t_i^-)\|$ . Then  $V(t_i^+) \leq (1 - \gamma)^2 V(t_i^-)$ .

953 This fits the Hespanha–Liberzon–Teel<sup>[46]</sup> (Theorem 1) framework for impulsive systems with desta-  
 954 bilizing continuous evolution and stabilizing impulses (the *reverse-ADT* branch). The continuous-  
 955 evolution rate is  $\lambda_c = \rho_\Sigma / R_\Sigma$  (from linearizing the disturbance contribution to  $\dot{V} \approx +\lambda_c V + \gamma_c(\rho_\Sigma)$ ).  
 956 The impulse contraction in  $V$  is  $\mu = (1 - \gamma)^2$ . The reverse-ADT condition for ultimate boundedness —  
 957 impulses cannot be too sparse — gives  $\Delta_{\max} \cdot \lambda_c < -\ln \mu \iff \Delta_{\max} \cdot \rho_\Sigma / R_\Sigma < -\ln(1 - \gamma)^2$ ,  
 958 where  $\Delta_{\max} := \sup_i (t_{i+1} - t_i)$  is the longest inter-update gap. Under the condition the modeled  
 959 mismatch is ultimately bounded within  $R_\Sigma^* \approx \rho_\Sigma \Delta_{\max} / \gamma$  (leading order in  $\gamma$ ).

960 Under uniform blocks  $\Delta_i = T/K$  the condition reduces to  $K/T > \rho_\Sigma / (2\gamma R_\Sigma)$  to leading order in  $\gamma$   
 961 — recovering ProST’s  $K/T$  form with the impulse gain made explicit. Under nonuniform schedules  
 962 the impulsive form is *strictly stronger*: the longest block sets the threshold, not the average. The  
 963 lemma is rigorous for the modeled impulsive sector dynamics; the per-update impulse gain  $\gamma$  is a  
 964 domain quantity (depends on within-block sample size and update-time discount) and is left abstract.  
 965  $\square$

#### 966 B.5 Proof of Key Lemma 3 (Loop generates interventional samples)

967 By temporal ordering and the singular-trajectory commitment of Section 3,  $a_t$  causally precedes  
 968  $o_{t+1}$ . We invoke Pearl’s do-calculus<sup>[35]</sup> (Theorem 3.4.1, Rule 2 — *action/observation exchange*):  
 969  $P(o_{t+1} \mid \text{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$  whenever  $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\overline{a_t}}}$ , where  $G_{\overline{a_t}}$  denotes  
 970 the graph  $G$  with all incoming arrows to  $a_t$  removed. In the singular-trajectory + agent-as-policy  
 971 graph of Section 3, the only arrows into  $a_t$  come from  $H_t$  (the agent’s policy is  $\pi_t(a_t \mid H_t)$ ); deleting  
 972 them severs the only confounding path from  $a_t$  to  $o_{t+1}$ . Whether  $H_t$  d-separates  $a_t$  from  $o_{t+1}$  in  $G_{\overline{a_t}}$   
 973 is precisely the content of (C2). The identity  $P(o_{t+1} \mid \text{do}(a_t), H_t) = P(o_{t+1} \mid a_t, H_t)$  holds on the  
 974 support where  $\pi_t(a_t \mid H_t) > 0$ , which is automatic since  $a_t$  was drawn from  $\pi_t$ .  $\square$

975 *Remark on equivalent forms of (C2).* The d-separation condition  $(o_{t+1} \perp\!\!\!\perp a_t \mid H_t)_{G_{\overline{a_t}}}$  is equivalent  
 976 to the potential-outcome form  $a_t \perp\!\!\!\perp Y^{(a_t)} \mid H_t$  (sequential ignorability in the Robins/Murphy sense),  
 977 and to the truncated-factorization condition  $P(\mathbf{U} \mid a_t, H_t) = P(\mathbf{U} \mid H_t)$  where  $\mathbf{U}$  are the latent  
 978 variables jointly affecting  $a_t$  and  $o_{t+1}$  — applied to the *prior* on  $\mathbf{U}$  in the do-marginalization, not to  
 979 the conditional on  $o_{t+1}$ . The three forms are interchangeable; the d-separation form is the most direct  
 980 handle for the proof above.

981 *On the role of (C1) and (C3).* The identification proof above is load-bearing on (C2) alone: (C1)’s full-  
 982 support positivity and (C3)’s known-action-mechanism are *automatic* under the paper’s architecture —  
 983 (C1) reduces to realized-action positivity ( $\pi_t(a_t \mid H_t) > 0$  tautologically since  $a_t$  was drawn from  $\pi_t$ );  
 984 (C3) reduces to agent-policy-queryability ( $Q_t$  is the agent’s own policy, internally accessible). Both  
 985 are stated as named conditions for forward-compatibility with off-policy / counterfactual extensions  
 986 where they would do real work; for the on-policy identification claim of this lemma, (C2) is the  
 987 substantive condition.

#### 988 B.6 Proof of Key Lemma 4 (Bias bound)

989 On the matching event  $\{a_{\text{ag},t}^*(s) = \tilde{a}_t^*(s)\}$ , the difference  $\hat{D}_t(s) - D_t^{\text{true}}(s)$  vanishes identically (both  
 990 quantities equal  $-\log Q_t$  at the same action). On the complementary misidentification event, the  
 991 support condition  $Q_t(\cdot \mid s) \geq q_0$  at both  $a_{\text{ag},t}^*(s)$  and  $\tilde{a}_t^*(s)$  implies  $\log Q_t(a_{\text{ag},t}^*(s) \mid s) \in [\log q_0, 0]$   
 992 and  $\log Q_t(\tilde{a}_t^*(s) \mid s) \in [\log q_0, 0]$ , so their absolute difference is at most  $\log(1/q_0)$ . Therefore  
 993  $|\hat{D}_t(s) - D_t^{\text{true}}(s)| \leq \mathbf{1}[a_{\text{ag},t}^*(s) \neq \tilde{a}_t^*(s)] \cdot \log(1/q_0)$ . Taking expectations:  $\mathbb{E}[|\hat{D}_t(s) - D_t^{\text{true}}(s)|] \leq$   
 994  $(1 - p_{\text{id}}(s)) \log(1/q_0)$  by definition of  $p_{\text{id}}(s)$ .  $\square$

995 **High-probability form** With probability  $1 - \delta$  over a single round’s identification event,  $|\hat{D}_t(s) -$   
996  $D_t^{\text{true}}(s)| \leq \log(1/q_0) \cdot \mathbf{1}[\text{misident}]$  where  $\Pr[\text{misident}] = 1 - p_{\text{id}}(s)$ . The bias decomposes  
997 clearly: in Regime A ( $p_{\text{id}}(s) \rightarrow 1$ ) it vanishes; in Regime B ( $p_{\text{id}}(s) \in (0, 1)$ ) it is controlled by  
998 misidentification mass; in Regime C ( $p_{\text{id}}(s) \rightarrow 0$ ) it saturates at  $\log(1/q_0)$ . This proves Key Lemma  
999 4.  $\square$

1000 **Per-state, per-horizon-step lift (used in Theorem 4.1(v)).** The lemma applies pointwise at any  
1001 visited state  $s_h$  along a horizon- $N_h$  trajectory:  $\mathbb{E}[|\hat{D}_t(s_h) - D_t^{\text{true}}(s_h)|] \leq (1 - p_{\text{id}}(s_h)) \log(1/q_0)$ ,  
1002 with  $p_{\text{id}}(s_h)$  the per-state argmax-correctness probability. Aggregating linearly over the horizon and  
1003 assuming the per-state floor  $p_{\text{id}} := \min_s p_{\text{id}}(s)$ , the per-round bias contribution to the trajectory-level  
1004 value gap is at most  $N_h(1 - p_{\text{id}}) \log(1/q_0)$ , contributing the second term of Theorem 4.1(v).

1005 **Remarks** - Both  $a_{\text{ag},t}^*(s)$  and  $\tilde{a}_t^*(s)$  must satisfy  $Q_t \geq q_0$  for the  $\log(1/q_0)$  ceiling to apply; the  
1006 support condition is on *both* points, not just the agent’s estimate. This is *strictly weaker* than the  
1007 perturbative identity’s full-support condition  $Q(a) \geq q_0$  for *all*  $a$  (Appendix B.2) — the perturbative  
1008 identity needs full support because  $\pi_e^*$  has positive mass on every action; the bias bound here needs  
1009 only two-point support. - This is *not* an estimation-from-samples lemma. If  $Q_t$  is unavailable internally  
1010 and must be estimated from on-policy rollouts, an additional concentration argument (Hoeffding  
1011 under iid fixed-policy samples, or martingale concentration for adaptive policies) is needed; in the  
1012 architectures this paper considers,  $Q_t$  is the agent’s policy and is internally available.

## 1013 C Auxiliary Mathematical Material

1014 This appendix collects supporting material used by the main results: convention hierarchy details, the  
1015 admissible-divergence family, direction-forcing argument, action-gap matching lower bound, tied-  
1016 optimum extensions, the structural-class theorem on gain-decay updates, strategic-tempo topologies,  
1017 deployment modes, and the chain-rule uniqueness theorem.

### 1018 C.1 Convention hierarchy: C1, C2, C3

1019 Three named continuation conventions form a hierarchy of increasing diagnostic power and computa-  
1020 tional cost.

1021 **C1 — One-step improvement (canonical default)**  $\pi_{\text{cont}} = \pi_{\text{current}}$ . Each action is evaluated  
1022 assuming current behavior continues afterward. No fixed-point computation, no global-optimality  
1023 assumption. Cheapest; weakest diagnostic.

1024 **C2 — Receding-horizon ( $N_r$ -step replanning)** At each future step, re-optimize  
1025 over a horizon of  $N_r$  steps using the model available at that step:  $\pi_{\text{RH}}(M_\tau) =$   
1026  $\arg \max_\pi V_O(M_\tau, \pi; N_r)$  applied at each  $\tau$ . Captures multi-step recovery: a goal that ap-  
1027 pears unattainable under frozen continuation may be reachable with replanning. Moderate cost;  
1028 moderate diagnostic.

1029 **C3 — Bellman**  $\pi_{\text{cont}} = \pi^*$  where  $\pi^* = \arg \max_\pi V_O(M_t, \pi; N_h)$ . The continuation is the optimal  
1030 policy — a fixed-point equation. Strongest diagnostic; most expensive.

1031 **Proposition C.1 (Monotonicity).** *For any model  $M_t$ , horizon  $N_h$ , and policy class  $\Pi$ :*  
1032  $A_O^{(1)}(M_t; \Pi, N_h) \leq A_O^{\text{RH}}(M_t; \Pi, N_r, N_h) \leq A_O^{\text{B}}(M_t; \Pi, N_h)$ .

1033 *Proof.* C1 freezes continuation at  $\pi_{\text{current}}$  (possibly suboptimal); C2 re-optimizes periodically, so  
1034  $\pi_{\text{RH}} \succeq \pi_{\text{current}}$  at each future step; C3 uses the globally optimal  $\pi^*$ . A weakly better continuation  
1035 yields a weakly higher expected trajectory value; taking the supremum over the first action preserves  
1036 the ordering.  $\square$

1037 **Corollary**  $\delta_{\text{sat}}^{\text{B}} \leq \delta_{\text{sat}}^{\text{RH}} \leq \delta_{\text{sat}}^{(1)}$  (since  $\delta_{\text{sat}} = V_O^{\min} - A_O$ , higher  $A_O$  means lower  $\delta_{\text{sat}}$ ). And  
1038  $\delta_{\text{regret}}^{(1)} \leq \delta_{\text{regret}}^{\text{RH}} \leq \delta_{\text{regret}}^{\text{B}}$  (since  $\delta_{\text{regret}} = A_O - V_O(M_t, \pi_{\text{current}}; N_h)$ , higher  $A_O$  means higher  $\delta_{\text{regret}}$ ).

C1 is the most conservative diagnostic (most likely to diagnose “locally unattainable”); C3 is the most accurate (least false “unattainable” diagnoses). The  $2 \times 2$  diagnostic structure is preserved under all three; only the inferential force varies.

## C.2 Admissible-divergence family for the regret bound

The reverse-KL direction is forced by the deterministic- $\pi^*$  regime (Appendix C.3). Within the direction-forced family, multiple smooth  $f$ -divergences yield valid regret bounds:

| Divergence                                          | Bound on $R(Q)$                            | Tightness               | Finite under det. $\pi^*$ ? |
|-----------------------------------------------------|--------------------------------------------|-------------------------|-----------------------------|
| $\text{TV}(\pi^*, Q)$                               | $V_{\max} \cdot \text{TV}$                 | Tight (extremal $V$ )   | Yes                         |
| $D_{\text{KL}}(\pi^* \  Q)$ via Pinsker             | $V_{\max} \sqrt{D_{\text{KL}}/2}$          | Loose by $\sqrt{\cdot}$ | Yes                         |
| $D_{\text{KL}}(\pi^* \  Q)$ via point-mass identity | $V_{\max}(1 - e^{-D_{\text{KL}}})$         | Tight (this paper)      | Yes                         |
| $\chi^2(\pi^* \  Q)$ (Le Cam)                       | $V_{\max} \cdot \frac{1}{2} \sqrt{\chi^2}$ | Looser than Pinsker     | $\chi^2 = 1/Q(a^*) - 1$     |
| $D_{\alpha}(\pi^* \  Q)$ (Rényi, $\alpha \geq 1$ )  | Various                                    | Interpolates            | Yes for $\alpha \geq 1$     |
| $D_{\text{KL}}(Q \  \pi^*)$ (forward-KL)            | $+\infty$                                  | Vacuous                 | No                          |

Reverse-KL is selected uniquely within the direction-forced family by the chain-rule axiom (Appendix C.10). The point-mass identity supplies the *exact* bound on reverse-KL under deterministic  $\pi^*$ ; the table reflects different bound shapes on the same divergence (with the BH inequality  $V_{\max} \sqrt{1 - e^{-D_{\text{KL}}}}$  as the looser general form to which the identity reduces outside the deterministic- $\pi^*$  corner).

## C.3 Direction-forcing argument

For deterministic  $\pi^* = \delta_{a^*}$  and any  $Q$  with  $Q(a) > 0$  for some  $a \neq a^*$ :  $D_{\text{KL}}(Q \| \pi^*) = \sum_a Q(a) \log \frac{Q(a)}{\pi^*(a)} = \sum_{a \neq a^*} Q(a) \log \frac{Q(a)}{0} = +\infty$ . A bound “ $R \leq +\infty$ ” is vacuous. The reverse direction  $D_{\text{KL}}(\pi^* \| Q)$  is finite (and equal to  $-\log Q(a^*)$ ) whenever  $Q(a^*) > 0$ . The asymmetry is forced by the regime: regret is a one-sided quantity (contributes only from  $Q$ ’s off-optimum mass;  $\pi^*$  has no support off  $a^*$ ); divergences whose role is to bound this one-sided quantity must themselves be one-sided. Symmetric divergences (squared Hellinger, Jensen-Shannon, symmetrized KL) introduce a vacuous symmetric term.

## C.4 Action-gap matching lower bound

For any  $Q$  with  $\Delta_{\min} = \min_{a \neq a^*} \Delta(a) > 0$ :  $R(Q) = \sum_{a \neq a^*} Q(a) \Delta(a) \geq \Delta_{\min} \sum_{a \neq a^*} Q(a) = \Delta_{\min} \cdot (1 - Q(a^*)) = \Delta_{\min} \cdot \text{TV}(\pi^*, Q)$ . By the point-mass identity (Lemma 5.1),  $\text{TV}(\pi^*, Q) = 1 - e^{-D_{\text{KL}}(\pi^* \| Q)}$ , giving the matching lower bound used in the two-sided characterization.

The lower bound is tight when sub-optimal actions are uniformly bad ( $\Delta_{\min} = \max_{a \neq a^*} \Delta(a)$ ). For typical landscapes the gap between upper and lower bound is  $V_{\max} - \Delta_{\min}$ , controlled by the *spread* of action gaps.

## C.5 Tied-optimum and softmax-smoothed extensions

**Tied-optimum** If  $\pi^*$  has support on a tied-optimum set  $\mathcal{A}^* = \{a : Q_O(a) = Q_O(a^*)\}$  with uniform mass, reverse-KL is finite whenever  $Q$  covers  $\mathcal{A}^*$ . The regret bound extends:  $R(Q) = \sum_{a \notin \mathcal{A}^*} Q(a) \Delta(a) \leq V_{\max} \cdot \mathbb{P}_Q(a \notin \mathcal{A}^*)$ . A multi-action analog of the point-mass identity holds with  $\pi^*(a^*) = 1/|\mathcal{A}^*|$  replacing the point-mass form, yielding  $D_{\text{KL}}(\pi^* \| Q) = -\sum_{a \in \mathcal{A}^*} (1/|\mathcal{A}^*|) \log(|\mathcal{A}^*|Q(a))$ , a multi-action analog of  $-\log Q(a^*)$ . The two-sided bound becomes one-sided in general; the upper bound holds with the modified KL form.

**Softmax-smoothed  $\pi^*$  with temperature  $\tau \rightarrow 0$**  Replace  $\pi^* = \delta_{a^*}$  with  $\pi_{\tau}^* \propto \exp(Q_O/\tau)$ . As  $\tau \rightarrow 0$ ,  $\pi_{\tau}^* \rightarrow \delta_{a^*}$ . For finite  $\tau$ , the *perturbative* identity (Appendix B.2) applies:  $\text{TV}(\pi_{\tau}^*, Q) = 1 - e^{-D_{\text{KL}}(\pi_{\tau}^* \| Q)} + O(\tau^{-1} \exp(-\Delta_{\min}/\tau))$  — exponentially small with polynomial prefactor. The two-sided regret bound transfers with the same correction order. Outside the perturbative regime — for genuinely high-entropy optima — the BH inequality  $\text{TV} \leq \sqrt{1 - e^{-D_{\text{KL}}}}$  becomes the relevant general bound; Pinsker also applies, with a one-sided regret form.

## 1079 C.6 The gain-decay structural class $\mathcal{A}_{\text{decay}}$

1080 The persistence inequality  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$  (Section 5.2) is an *instantaneous* check at the current  
 1081 operating point. Define the structural class by gain-decay rather than sample-counting:  $\mathcal{A}_{\text{decay}} :=$   
 1082  $\{\text{updates whose effective sector gain } \alpha_{ij}^{(t)} \rightarrow 0 \text{ as } t \rightarrow \infty \text{ on every revisable element}\}.$

1083 **Theorem C.2** (Universal failure of  $\mathcal{A}_{\text{decay}}$ ). *For any fixed  $(\rho_{\Sigma}, R_{\Sigma})$  with  $\rho_{\Sigma} > 0$ , every agent in*  
 1084  $\mathcal{A}_{\text{decay}}$  *eventually violates the persistence threshold  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} > \rho_{\Sigma}/R_{\Sigma}$ .*

1085 *Proof.* At every element  $(i, j)$ ,  $\alpha_{ij}^{(t)} \rightarrow 0$  by definition of  $\mathcal{A}_{\text{decay}}$ , so  $\min_{(i,j)} \alpha_{ij}^{(t)} \rightarrow 0$ , so the  
 1086 bottleneck  $\mathcal{T}_{\Sigma}^{\text{bn}} \rightarrow 0$  with experience. Once  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} < \rho_{\Sigma}/R_{\Sigma}$ , the threshold is violated. This holds  
 1087 *regardless of prior calibration*: at any finite calibration level, the gain decays below threshold once  
 1088 enough experience accumulates.  $\square$

1089 *Scope clarification (per-element pull).* The class definition requires  $\alpha_{ij}^{(t)} \rightarrow 0$  on *every* revisable  
 1090 element — i.e., the agent is being pulled at every element repeatedly. Beta-Bernoulli with  $\eta_{\text{edge},ij} =$   
 1091  $1/(n_{ij} + 1)$  is in the class only for elements actually being updated; an agent that ignores a subset of  
 1092 elements is outside the class for those elements (their gain remains at initial calibration), but inside  
 1093 it for the elements it does touch. The theorem applies to the touched elements; persistence on the  
 1094 untouched ones reduces to whatever calibration was set initially.

1095 This is a *structural* failure of the class, not a tuning problem. The class includes: - Count-accumulating  
 1096 Bayesian updates without forgetting (e.g., Beta-Bernoulli with  $\eta_{\text{edge},ij} = 1/(n_{ij} + 1)$  where  $n_{ij} \rightarrow \infty$ ).  
 1097 - Bounded-memory schemes with growing memory. - Observation-aggregating schemes without restart.  
 1098 - Gradient-based methods with vanishing step sizes ( $\eta_t = c/t^{\beta}$  for  $\beta \in (0, 1]$ , the Robbins–Monro  
 1099 regime).

1100 Mechanisms *outside*  $\mathcal{A}_{\text{decay}}$  — constant-step-size stochastic approximation, sliding-window updates,  
 1101 bounded-memory learners, block-restart schemes, Kalman filters with bounded process noise —  
 1102 maintain a finite gain ceiling and escape asymptotic decay, but then face a *bidirectional* threshold:

Table 1: Bidirectional thresholds for non-accumulating mechanisms outside  $\mathcal{A}_{\text{decay}}$ .

| Mechanism                                                                                                        | $\alpha_{\Sigma}^{\text{ss}}$ | Prerequisite holds iff                            |
|------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------|
| Exponential forgetting with $\lambda$                                                                            | $1 - \lambda$                 | $1 - \lambda > \rho_{\Sigma}/R_{\Sigma}$          |
| Constant-step SA with rate $\eta$                                                                                | $\eta$                        | $\eta > \rho_{\Sigma}/R_{\Sigma}$                 |
| Fixed-window mechanisms (sliding-<br>window size $W$ , bounded-memory<br>size $K$ , block-restart period $T_R$ ) | $1/W, 1/K, 1/T_R$             | window/memory/period $< R_{\Sigma}/\rho_{\Sigma}$ |

1103 For each row, the threshold is *bidirectional* and sharp within the sector-Lyapunov reduction.

## 1104 C.7 Strategic-tempo across canonical topologies

1105 The bottleneck strategic tempo  $\mathcal{T}_{\Sigma}^{\text{bn,ss}}$  of Lemma 5.2 is verified — and the resulting per-topology  
 1106 threshold derived — across four canonical topologies under Beta-Bernoulli edge updates with per-  
 1107 element forgetting:

- 1108 • **(T1) Single edge.**  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \nu \cdot \iota \cdot (1 - \lambda)$ . All three factors load-bearing; setting any one to  
 1109 zero collapses the bottleneck.
- 1110 • **(T2) Two-edge AND chain, observable intermediate.**  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min(\nu_1(1 - \lambda_1), \nu_1\theta_1(1 -$   
 1111  $\lambda_2))$ . Edge 2’s effective observation rate is gated by edge 1’s success probability  $\theta_1$  —  
 1112 *depth-gated attenuation*. For depth- $d$  chains, the bottleneck is  $\min_k \prod_{j < k} \theta_j(1 - \lambda_k)$ .
- 1113 • **(T3) Two-edge AND chain, unobservable intermediate.** Per-edge tempo ill-defined; plan-  
 1114 level bottleneck  $\mathcal{T}_{\Sigma, \text{plan}}^{\text{bn,ss}} = \nu(1 - \lambda_{\Phi})$  over a single tracked plan-level quantity  $\hat{\Phi} = p_1 p_2$ .

- **(T4) Two-arm OR node,  $\varepsilon$ -greedy.**  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min((1 - \varepsilon)(1 - \lambda_1), \varepsilon(1 - \lambda_2))$ . Action selection controls rate allocation; pure greedy ( $\varepsilon = 0$ ) collapses the unexplored arm's bottleneck — *exploration-gated*.

The structural decomposition: AND-chains exhibit *depth-gated* geometric attenuation  $\nu_k = \nu \prod_{j < k} \theta_j$ ; OR-nodes exhibit *exploration-gated* allocation. Mixed AND/OR DAGs interleave both. Each topology surfaces a different factor as the binding bottleneck.

**Regime-C contamination is structurally fatal**  $\nu_{ij} = 0$  at any single element  $\Rightarrow \mathcal{T}_{\Sigma}^{\text{bn,ss}} = 0$ . An agent with even one fully-confounded element cannot meet the threshold by forgetting alone.

## C.8 Three deployment modes of loop-Level-2

The Pearl-do structure of closed-loop interventional access manifests at distinct layers with semantically distinct interventional mechanisms:

- **Mode 1 — agent-self-intervention.** The agent performs do-actions on its own action space as part of its ordinary adaptive loop. Intervention is on the agent's own action; target is the environment's response. This is Lemma 5.3's primary content.
- **Mode 2 — observer-on-sub-agent.** An observer external to a composite performs do-interventions on one sub-agent's action space; target is another sub-agent's response. Reveals cross-coupling structure that component-marginal observation cannot identify.
- **Mode 3 — observer-on-agent-input** (sketched, future work). An observer intervenes on the agent's observation channel; target is the agent's subsequent policy. Useful for offline architectural-class diagnosis.

Modes share the Pearl-do structure but differ in who intervenes on what. The unification is at the pattern level; the mechanism is semantically distinct per layer.

## C.9 Distinction from active inference and causal-RL precursors

Action-perception-loop frameworks — active inference<sup>[30,31]</sup>, control-as-inference<sup>[48]</sup>, cybernetics<sup>[49,50]</sup> — implicitly use the action-causes-observation observation. Our distinctive moves:

- *Bareinboim-hierarchy connection.* Active inference / cybernetics rest on Bayesian-network (Level 1) generative models; we invoke the causal-hierarchy theorem<sup>[29]</sup> to position the policy DAG as causal with do-conditioning in  $Q_O$ .
- *Regime-indexed identifiability (A/B/C).* Active inference literature treats causal identifiability uniformly within modeling assumptions; we surface the regime split at framework level.
- *Scope honesty.* We distinguish “data generated under intervention” from “cleanly identified do-estimates”;<sup>[47]</sup> documents that the active-inference literature sometimes elides this.

The causal-RL line<sup>[13–18]</sup> is the direct ancestor for regime-indexed identifiability and on-policy interventional access; all are stationary-MDP. Composition with non-stationarity is, to our knowledge, novel.

## C.10 Chain-rule uniqueness of reverse-KL

**Theorem C.3** (Chain-rule uniqueness (Hobson 1969; Csiszár 1991)). *Let  $D_f(P \| Q) = \sum_x Q(x) f(P(x)/Q(x))$  be a smooth  $f$ -divergence with  $f$  convex and  $f(1) = 0$ . The chain rule  $D_f(P_{XY} \| Q_{XY}) = D_f(P_X \| Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \| Q_{Y|X})]$  holds for all joint distributions if and only if  $f(t) = c \cdot t \log t$  for some  $c > 0$  — i.e.,  $D_f$  is reverse-KL up to positive scaling.*

*Proof.* Take any joint  $P_{XY}, Q_{XY}$  on a finite product space. Writing  $r_x := P(x)/Q(x)$  and  $s_{y|x} := P(y | x)/Q(y | x)$ , the joint ratio factorizes as  $P_{XY}(x, y)/Q_{XY}(x, y) = r_x \cdot s_{y|x}$ . Expanding both sides of the chain rule:  $D_f(P_{XY} \| Q_{XY}) = \sum_{x,y} Q_{XY}(x, y) f(r_x s_{y|x}) = \sum_x Q(x) \sum_y Q(y | x) f(r_x s_{y|x})$ ,  $D_f(P_X \| Q_X) + \mathbb{E}_{P_X}[D_f(P_{Y|X} \| Q_{Y|X})] = \sum_x Q(x) f(r_x) + \sum_x P(x) \sum_y Q(y | x) f(s_{y|x})$ .

1159  $x) f(s_{y|x})$ , the second term using  $\mathbb{E}_{P_X} = \sum_x P(x) = \sum_x Q(x) r_x$ . Equating and collecting:  
1160  $\sum_x Q(x) \left[ \sum_y Q(y | x) f(r_x s_{y|x}) - f(r_x) - r_x \sum_y Q(y | x) f(s_{y|x}) \right] = 0$ . The chain rule  
1161 must hold for *all* joints, including those where  $r_x$  takes arbitrary positive value at one  $x$  and  $s_{y|x}$   
1162 varies independently — which forces the bracketed expression to vanish pointwise. A two-point  
1163 reduction (take  $Y$  binary with  $s_{y|x}$  taking values  $s$  and  $1/s$  on equal-weight  $Q(y | x)$ ) collapses  
1164 the inner sums and yields the functional equation  $f(rs) = f(r) + r f(s)$  for all  $r, s > 0$ .  
1165 With  $f$  convex and  $f(1) = 0$ , the unique solution is  $f(t) = c \cdot t \log t$  for  $c > 0$ <sup>[51]</sup> (§4) — this  
1166 gives  $D_f(P \| Q) = c \sum_x Q(x) (P(x)/Q(x)) \log(P(x)/Q(x)) = c \sum_x P(x) \log(P(x)/Q(x)) =$   
1167  $c \cdot D_{\text{KL}}(P \| Q)$ , reverse-KL up to positive scaling.  $\square$

1168 **References** Hobson 1969 (“A new theorem of information theory,” *J. Stat. Phys.*); Csiszár 1991  
1169 (“Why least squares and maximum entropy?” *Annals of Statistics*; Theorem 3 corollary, Theorem 5);  
1170 Shore-Johnson 1980 (“Axiomatic derivation of the principle of maximum entropy,” *IEEE Trans. Info.*  
1171 *Theory*, system-independence axiom); Sanov 1957 (large-deviation rate function); Aczél-Daróczy  
1172 1975 (functional-equation machinery).

1173 These references give *structurally equivalent reformulations* of the same axiom. The Cauchy functional  
1174 equation each reduces to is the common content. No known uniqueness route outside the independence-  
1175 on-sub-problems family exists.

1176 **Why other family members fail the chain rule** Concrete counterexample for  $\chi^2$ : take  $Q_X$  uniform  
1177 on  $\{x_1, x_2\}$ ,  $P_X = (3/4, 1/4)$ ,  $Q(y|x)$  uniform,  $P(y|x) = (3/4, 1/4)$ . Direct calculation gives  
1178  $\chi^2(P_{XY} \| Q_{XY}) = 9/16$  while  $\chi^2(P_X \| Q_X) + \mathbb{E}_{P_X} [\chi^2(P_{Y|X} \| Q_{Y|X})] = 1/4 + 1/4 = 8/16$ .  
1179 Non-additive. Rényi- $\alpha$  for  $\alpha \neq 1$  fails analogously; squared Hellinger likewise fails.

## 1180 D Algorithm Sketch and Base-Learner Instantiation

1181 A practical algorithm achieving the joint guarantees of Theorem 4.1 requires four components, each  
1182 translating one of the assumptions (A1)–(A5) into operational form.

1183 **(a) Identity-tight base learner satisfying (A5)** In the stationary deterministic- $\pi^*$  bandit case ( $N_h =$   
1184  $1$ ), Thompson sampling<sup>[19]</sup> (posterior-randomized; pointwise  $Q_t(a^*) > 0$ ) and vanilla deterministic  
1185 UCB<sup>[27]</sup> (Dirac-deployed;  $Q_t(a^*) \in \{0, 1\}$ , with the (A1) extended-real reading of Theorem 4.1) both  
1186 achieve the per-round coordinate in expectation:  $\mathbb{E}[1 - e^{-K_t}] = \mathbb{E}[1 - Q_t(a^*)] = O\left(\frac{\log t}{t \Delta_{\min}^2}\right)$ ,  
1187 following from Lattimore and Szepesvári 2020<sup>[27]</sup> Theorem 7.1 ( $\mathbb{E}[N_a(t)] \leq 16 \log(t)/\Delta_a^2 + O(1)$   
1188 per suboptimal arm; aggregating over arms with the per-state floor gives the per-round form). This  
1189 satisfies (A5) of Theorem 4.1 with the *stronger logarithmic* rate, yielding cumulative dynamic  
1190 regret  $O(V_{\max}(B_T + 1) \log(T/(B_T + 1))/\Delta_{\min}^2)$ . The  $V_{\max} \cdot \text{TV}$  chain is structurally one factor of  
1191  $\Delta_{\min}$  looser than direct gap-aware UCB analysis (which gives  $O(\log T/\Delta_{\min})$  cumulative regret per  
1192 block by weighting each suboptimal pull by its own  $\Delta(a)$  rather than  $V_{\max}$ ); this is a feature of the  
1193 unification — the framework’s contribution is composition + non-stationarity-handling + identity-  
1194 coordinate exactness, not a sharper bandit constant. For  $N_h > 1$  MDPs, UCRL2<sup>[52]</sup> and UCBVI<sup>[43]</sup>  
1195 give per-block cumulative trajectory-level regret  $\tilde{O}(N_h^{3/2} \sqrt{SAL})$  with  $S, A$  the state/action sizes;  
1196 rewriting in the occupancy-weighted coordinate satisfies (A5) with  $c$  of order  $\tilde{O}(\sqrt{N_h SA})$ , lifting  
1197 cleanly to the  $\tilde{O}(N_h^2 \sqrt{SA(B_T + 1)T})$  cumulative rate — same piecewise-stationary  $B_T$  family as  
1198 restart-on-change variants of<sup>[11]</sup> (the continuous-variation regime of<sup>[3]</sup> is structurally distinct; see  
1199 Section 4 *Comparator regime*). The occupancy-weighted form is the natural object for trajectory-  
1200 level base learners; the per-state KL/TV identity is preserved underneath. Information-directed  
1201 sampling<sup>[20]</sup> requires a different analysis here because  $H(\pi^*) = 0$  collapses its information ratio at  
1202 the deterministic- $\pi^*$  corner; we leave the IDS analysis as future work.

1203 **(b) Multi-factor forgetting schedule satisfying (A2)** Choose per-element discount rates  $\{\lambda_{ij}\}$   
1204 such that the bottleneck  $\mathcal{T}_{\Sigma}^{\text{bn,ss}} = \min_{(i,j)} \nu_{ij} \iota_{ij} (1 - \lambda_{ij})$  exceeds  $\rho_{\Sigma}/R_{\Sigma}$ . When  $\rho_{\Sigma}$  is unknown,  
1205 the<sup>[2]</sup> black-box reduction gives an adaptive choice (applied per element if heterogeneous); when  $\rho_{\Sigma}$

is known via a variation-budget<sup>[1]</sup> or change-point detection<sup>[39]</sup>, the choice is direct. As a fail-fast pre-check,  $\mathcal{T}_{\Sigma}^{\text{agg,ss}} > |E| \cdot \rho_{\Sigma}/R_{\Sigma}$  is necessary.

**(c) Identifiability check satisfying (A3)** Test whether the loop data is in Regime A (full identifiability), Regime B (partial), or Regime C (none). In Regime A, no adjustment is needed. In Regime B, apply<sup>[17]</sup> (DOVI)-style confounding-bias adjustment. In Regime C, the bound is provable but not realizable; algorithm flags the regime as out-of-scope.

**(d) Two-gap diagnostic readout satisfying (A4)** Compute  $\delta_{\text{sat}}$  (requires  $A_O$ , intractable in general; estimable via the policy-class supremum over recent rounds) and  $\delta_{\text{regret}}$  (requires  $V_O$  at current policy; tractable in simulation). Route to corrective action class.

The algorithm is sketch-grade; full instantiation and empirical evaluation are deferred to a follow-up paper.

## E Pinsker vs. Point-Mass Identity Numerical Comparison

We compare  $V_{\max}(1 - e^{-D_{\text{KL}}})$  (point-mass identity bound, exact under deterministic  $\pi^*$  per Lemma 5.1 and the two-sided regret theorem) with  $V_{\max}\sqrt{D_{\text{KL}}/2}$  (Pinsker bound, also valid under deterministic  $\pi^*$  but loose),  $V_{\max}\sqrt{1 - e^{-D_{\text{KL}}}}$  (Bretagnolle–Huber inequality, the general envelope below which the identity sits at this corner), and the trivial envelope  $V_{\max}$ . Set  $V_{\max} = 1$  for normalization.

Table 2: Identity vs. Pinsker vs. BH at  $V_{\max} = 1$ .

| $D_{\text{KL}}$ | $1 - e^{-D_{\text{KL}}}$ (identity) | $\sqrt{1 - e^{-D_{\text{KL}}}}$ (BH) | $\sqrt{D_{\text{KL}}/2}$ (Pinsker) | ratio |
|-----------------|-------------------------------------|--------------------------------------|------------------------------------|-------|
| 0.01            | 0.00995                             | 0.0998                               | 0.0707                             | 7.10× |
| 0.1             | 0.0952                              | 0.308                                | 0.224                              | 2.35× |
| 0.5             | 0.393                               | 0.627                                | 0.500                              | 1.27× |
| 1.0             | 0.632                               | 0.795                                | 0.707                              | 1.12× |
| 2.0             | 0.865                               | 0.930                                | 1.000                              | 1.16× |
| 4.0             | 0.982                               | 0.991                                | 1.414                              | 1.02× |
| 10.0            | 0.99995                             | 0.99998                              | 2.236                              | 1.00× |

The point-mass identity bound is uniformly tighter than Pinsker, by a factor of  $7\times$  at  $D_{\text{KL}} = 0.01$  and converging to 1 as  $D_{\text{KL}}$  grows large (where both saturate at  $V_{\max}$ ). It is also uniformly tighter than the BH inequality  $\sqrt{1 - e^{-D_{\text{KL}}}}$  — the relation  $x < \sqrt{x}$  on  $(0, 1)$  — although both saturate together at  $V_{\max}$  for large  $D_{\text{KL}}$ . Pinsker, by contrast, hits the trivial envelope  $V_{\max} = 1$  at  $D_{\text{KL}} = 2$  and exceeds it beyond — vacuous from  $D_{\text{KL}} = 4$  onward (where the unclipped form returns 1.414), fully vacuous by  $D_{\text{KL}} = 10$ .

The matching lower bound  $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$  has the same shape, scaled by  $\Delta_{\min}/V_{\max}$ . In the extremal value landscape ( $\Delta_{\min} = V_{\max}$ ), the upper and lower bounds coincide and the regret is *identified exactly* by the KL coordinate. For typical landscapes, regret and KL are Lipschitz-equivalent with constants  $\Delta_{\min}/V_{\max}$  and 1.

## F Prior-Art Retrieval Summary

We conducted a structured prior-art retrieval (via Undermind) with the goal of identifying any framework composing all four target elements: (1) goal-feasibility-vs-policy-quality two-gap diagnostic; (2) an exact reverse-KL/TV regret identity (point-mass) under deterministic- $\pi^*$ , or — more loosely — Bretagnolle–Huber-style use in regret analysis; (3) strategic-tempo / forgetting prerequisite tying convergence to rate of useful policy revision; (4) closed-loop interventional access making regret bounds learnable from on-policy data.

**Result: 63 papers retrieved, abstract-level coverage estimated 75%, no direct anticipation in the retrieved set** The landscape splits into four largely separate strands corresponding to our four components; we did not find a published framework composing all four in this retrieval. Retrieval

1242 queries used the structured prior-art tool Undermind; the four-strand cite-and-distinguish in Section 2  
1243 surfaces the closest neighbor in each strand.

1244 **Strongest negative signal (in the retrieved corpus): the Bretagnolle–Huber inequality is absent**  
1245 **as an upper-bound coordinate for regret analysis in the retrieved RL / non-stationary literature**

1246 The information-theoretic regret line —<sup>[19–24]</sup> — uses entropy / mutual information / information  
1247 ratio / Pinsker / Hellinger; we did not find the BH inequality deployed *as an upper-bound regret*  
1248 *coordinate*. (BH is well-known in bandit *lower-bound* proofs via Le Cam’s method and Tsybakov’s  
1249 framework<sup>[26]</sup>, including in standard RL textbook treatments<sup>[27]</sup>; that use is not contested here.) The  
1250 *exact point-mass identity at the deterministic- $\pi$  corner\** (Lemma 5.1) — which strictly improves both  
1251 Pinsker and the BH envelope at this corner — was likewise not found in the retrieved corpus, in either  
1252 upper-bound or lower-bound use. We frame this as a literature-search observation against an explicitly  
1253 bounded retrieval (63 papers, ~75% abstract-level coverage), not a formal absence claim against all  
1254 published RL work; reviewers should read it as “to our knowledge / in the retrieved set” rather than  
1255 “absolutely novel.”

1256 **Closest neighbors per strand:**<sup>[1]</sup> (Strand 1 — variation-budget dynamic regret);<sup>[5]</sup> (Strand 2 —  
1257 closest two-term decomposition, different axis);<sup>[8]</sup> (Strand 3 — closest tempo result, different form);<sup>[14]</sup>  
1258 (Strand 4 — closest causal-RL, stationary only).

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1266 factor strategic tempo with forgetting prerequisite, closed-loop interventional access) and  
1267 the cumulative dynamic regret rate. Each component is developed in the main-result and  
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